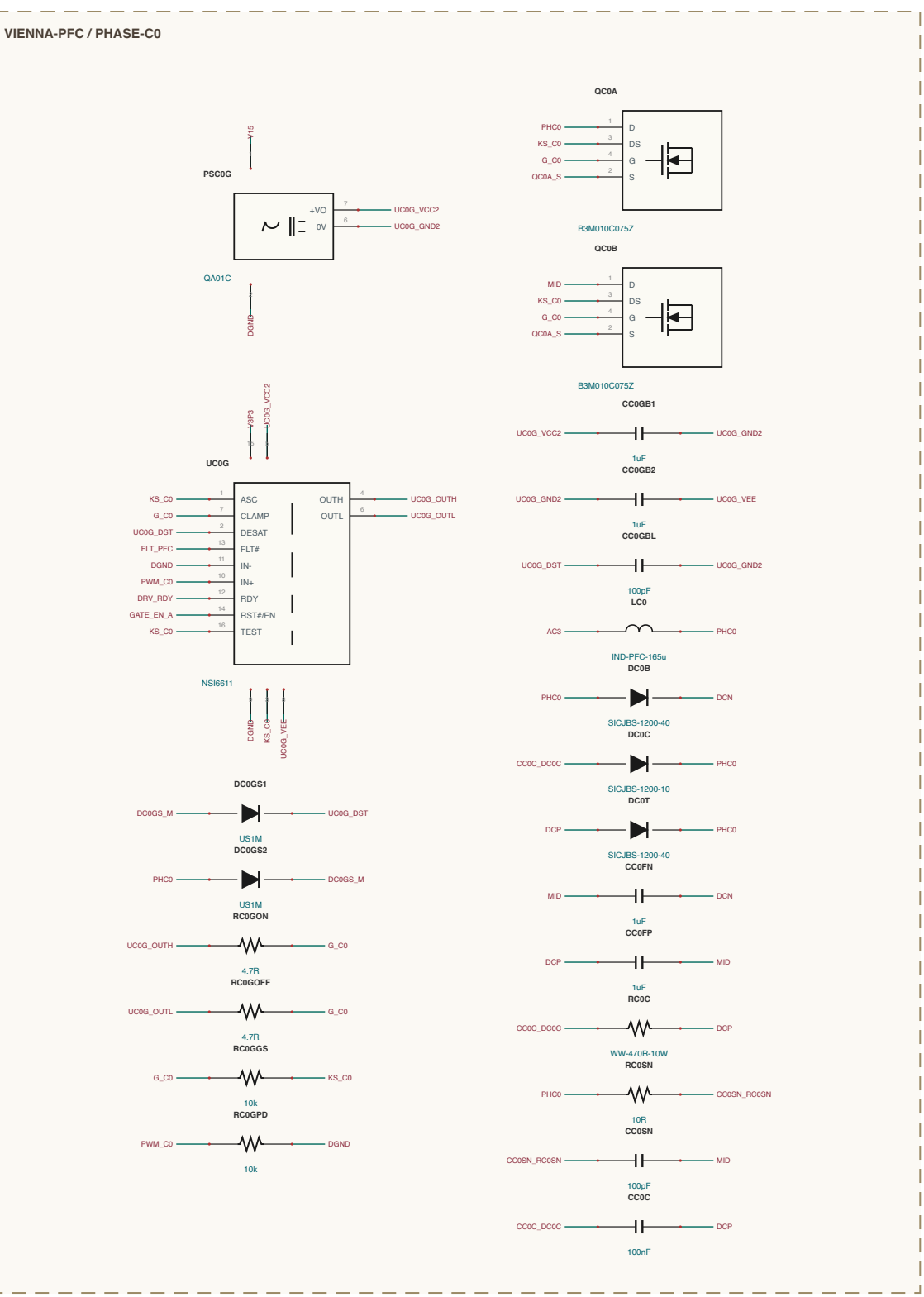
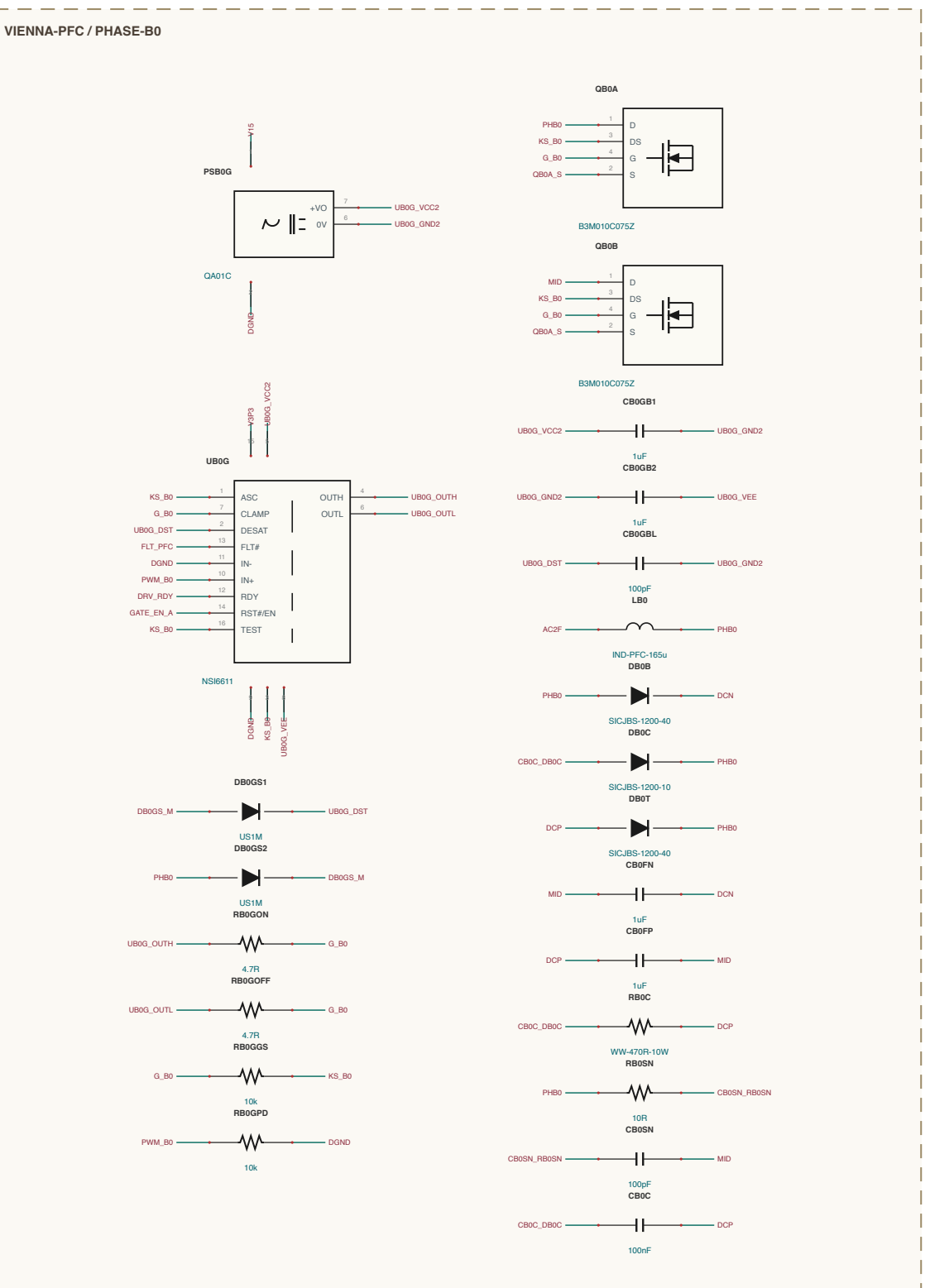
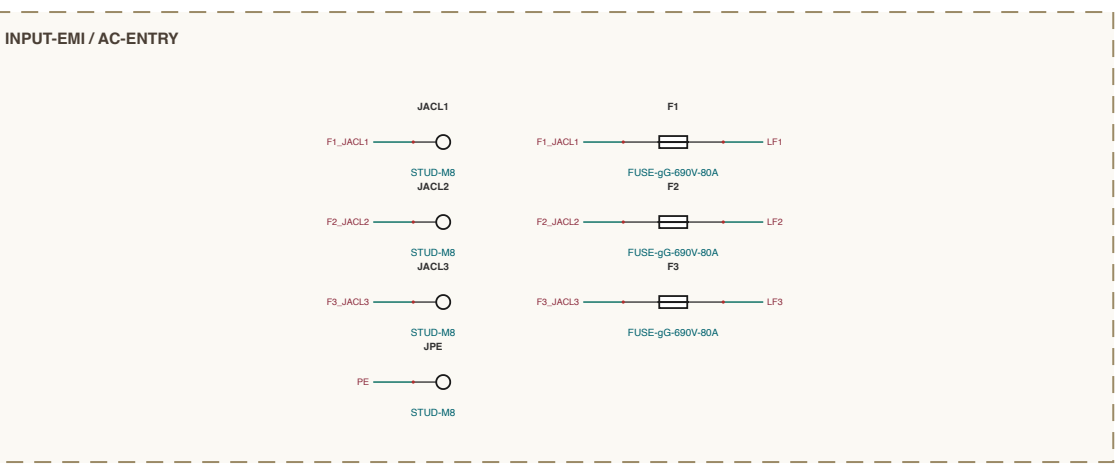
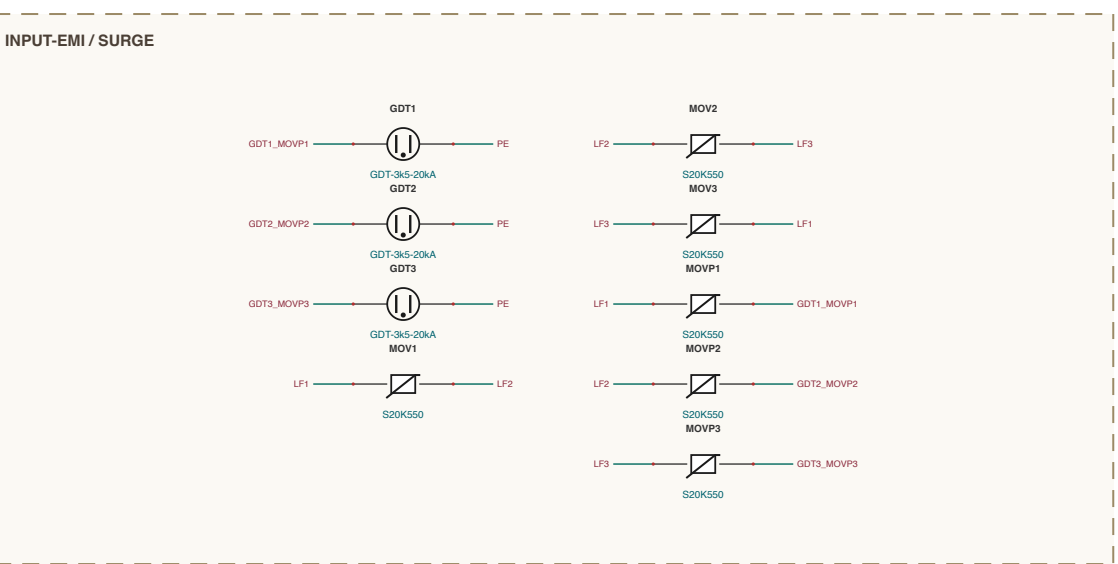
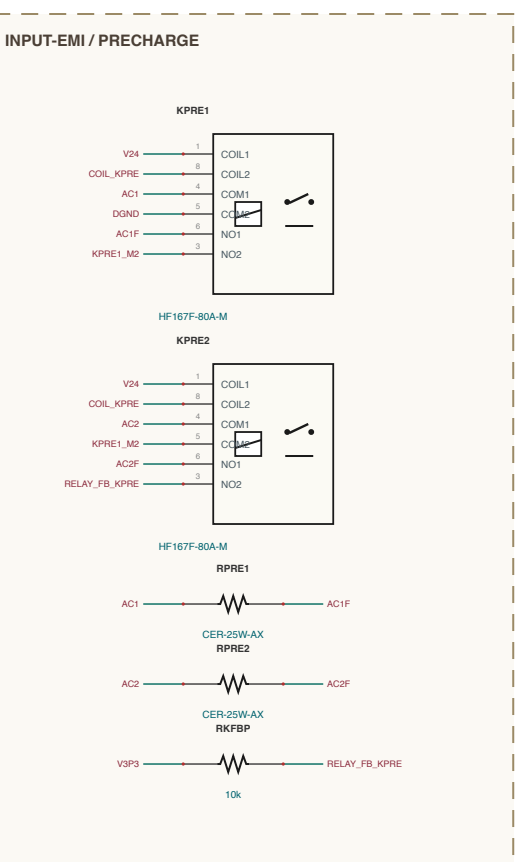
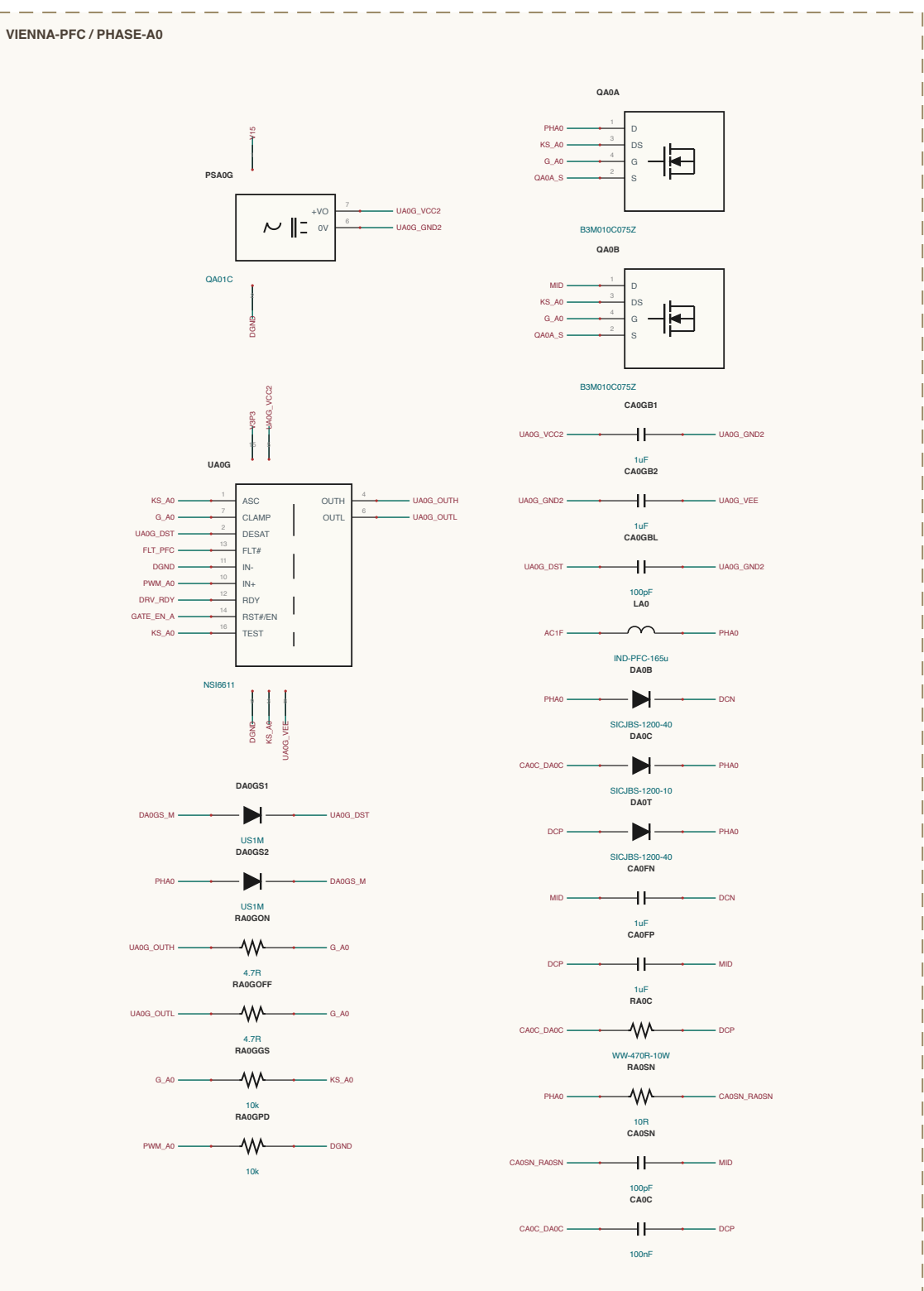
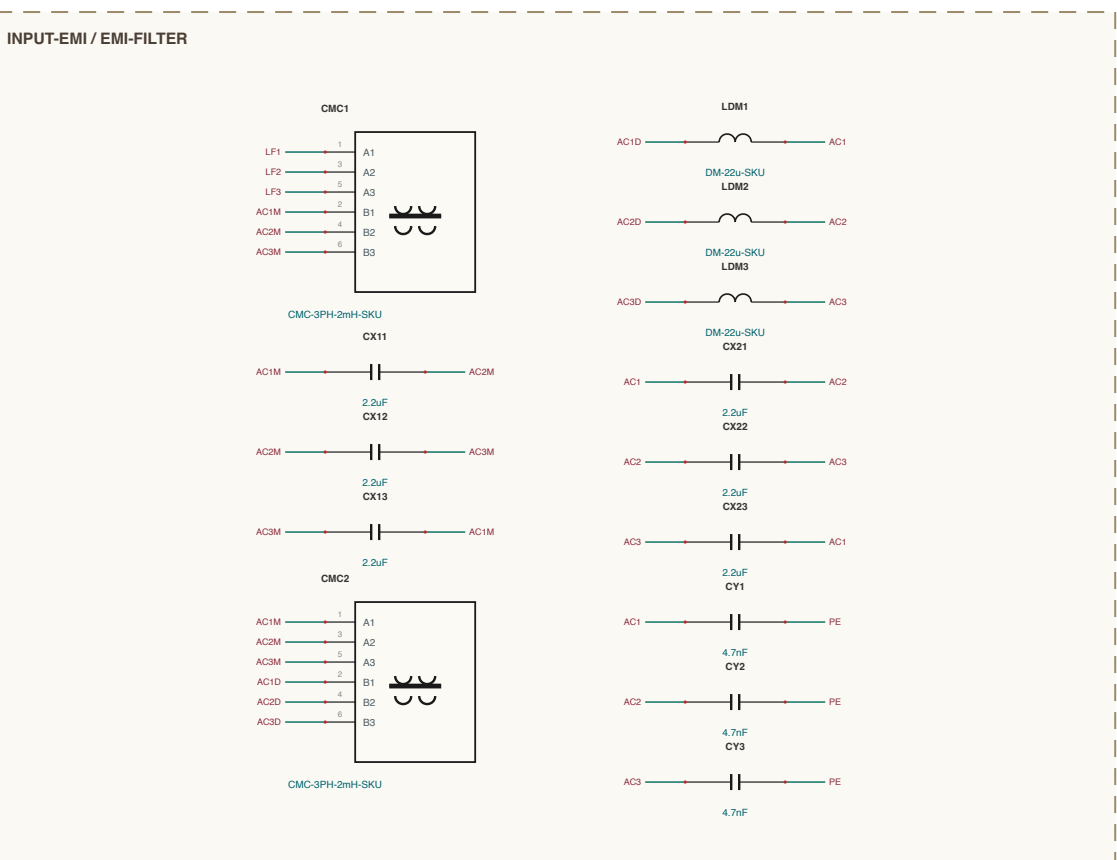
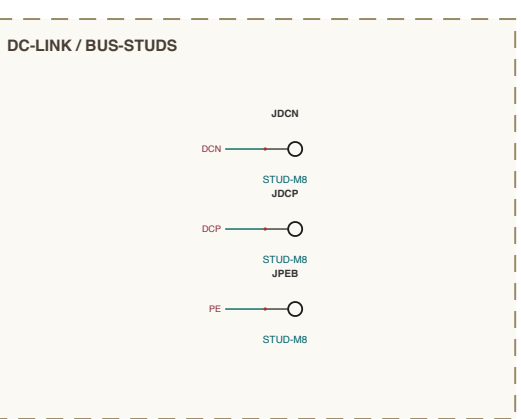
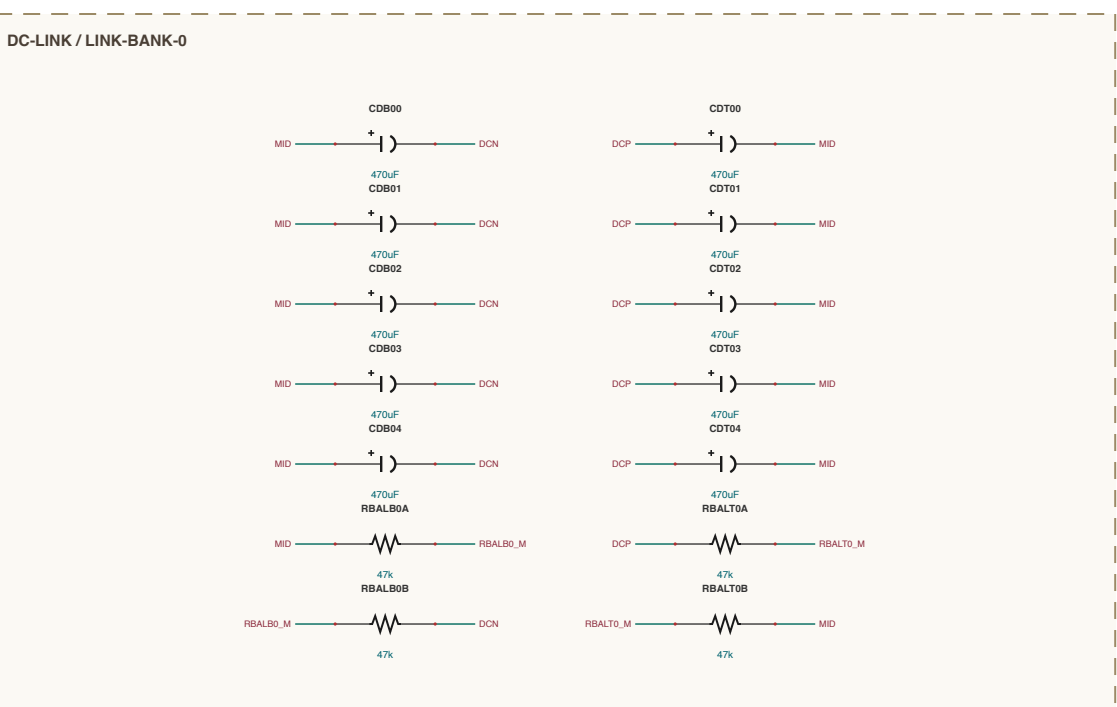
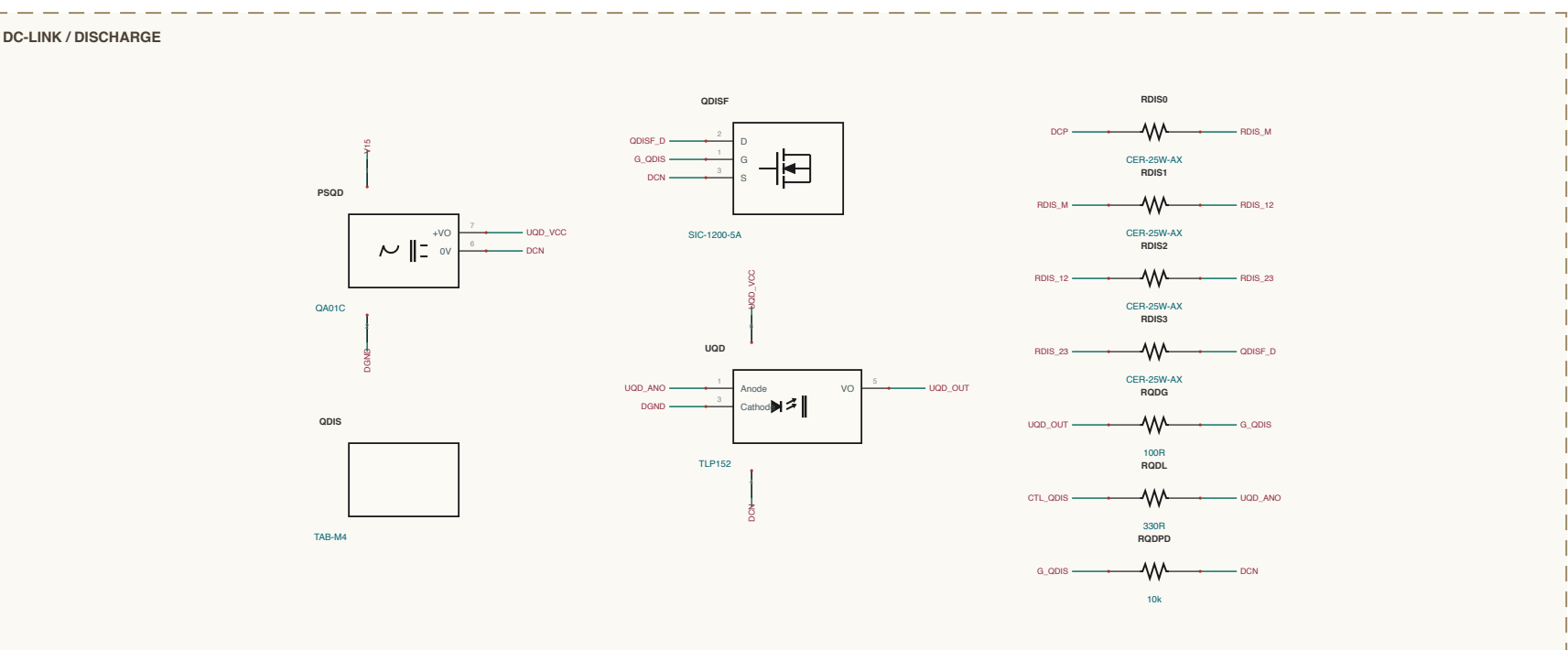
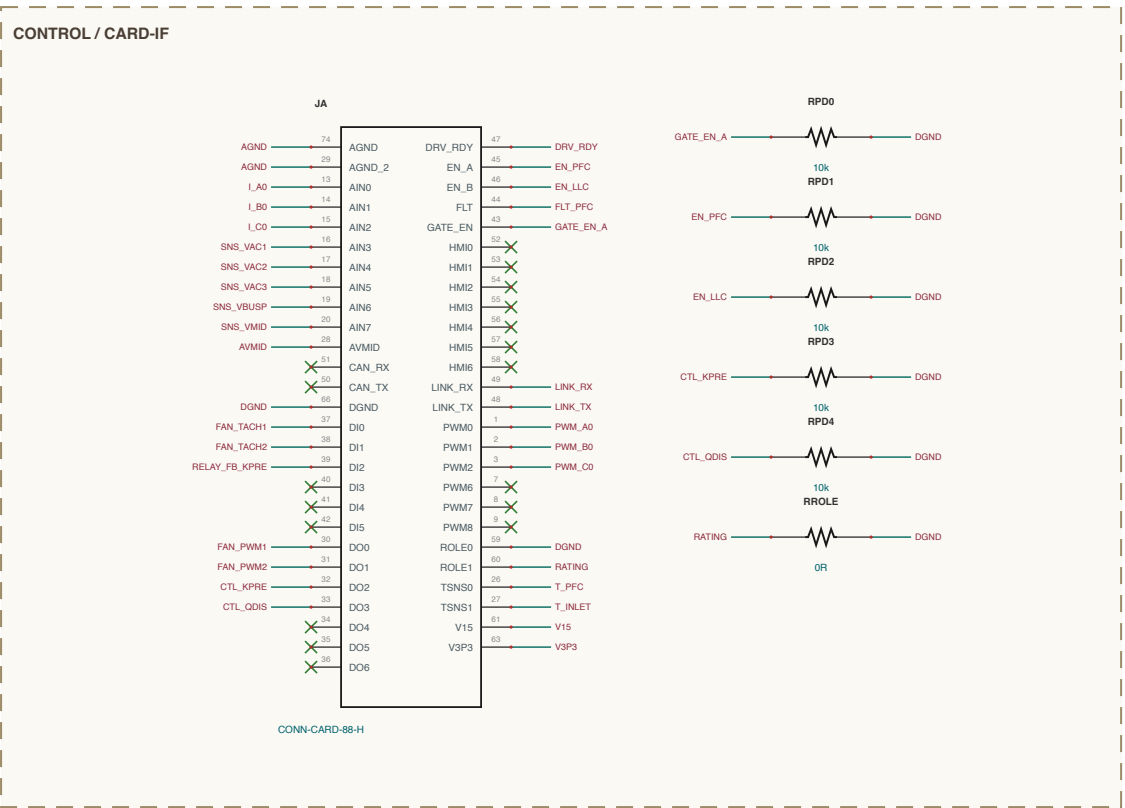
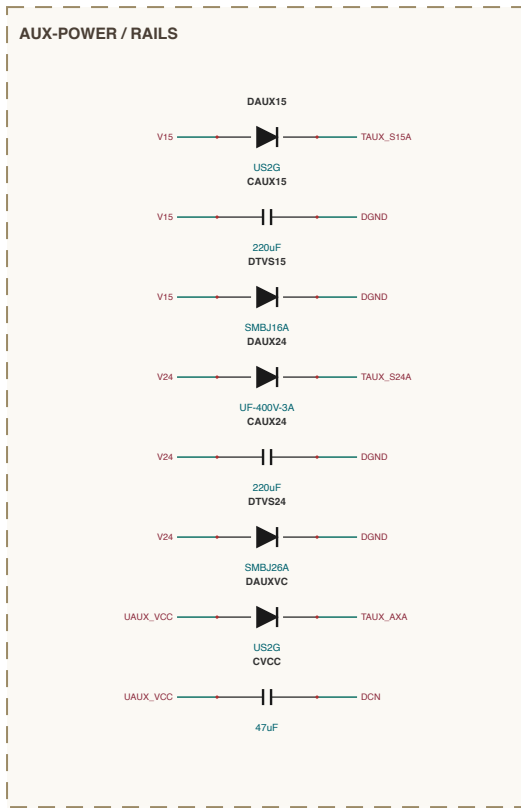
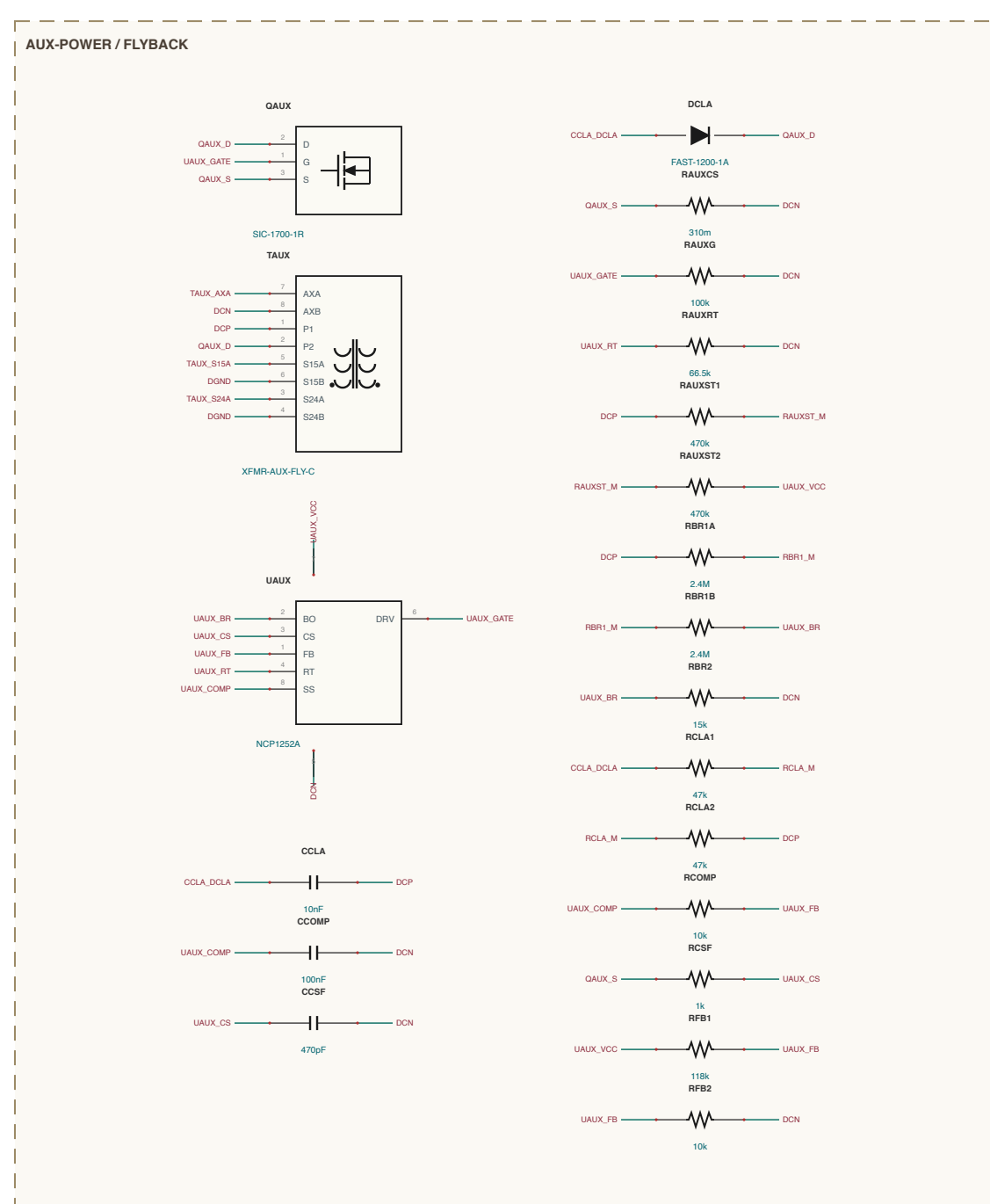
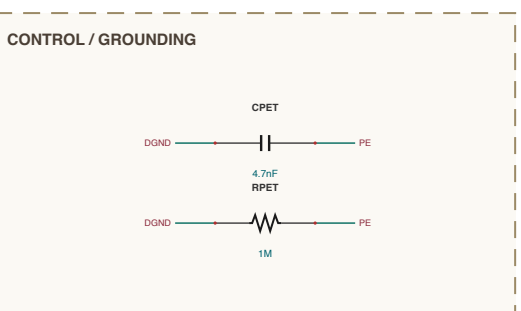
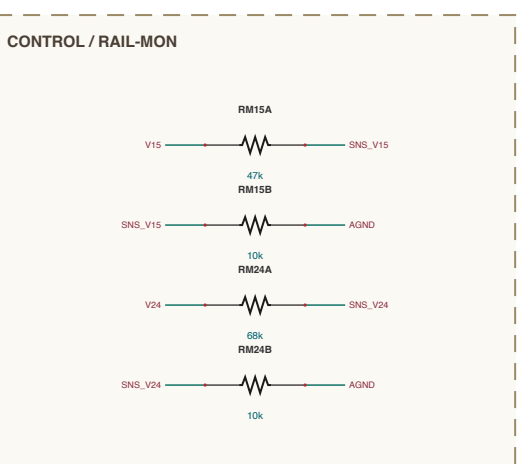
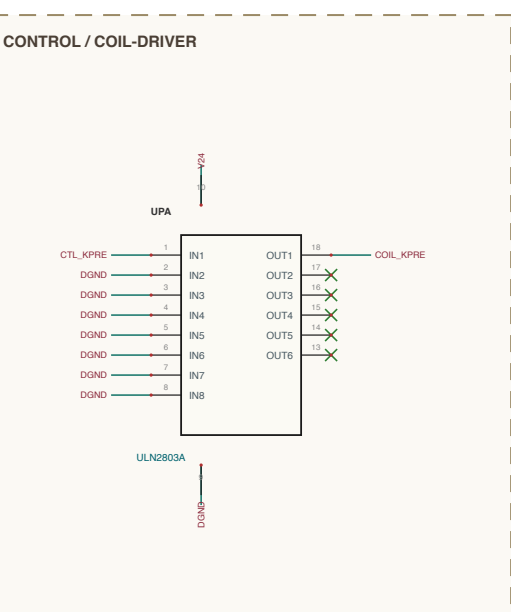
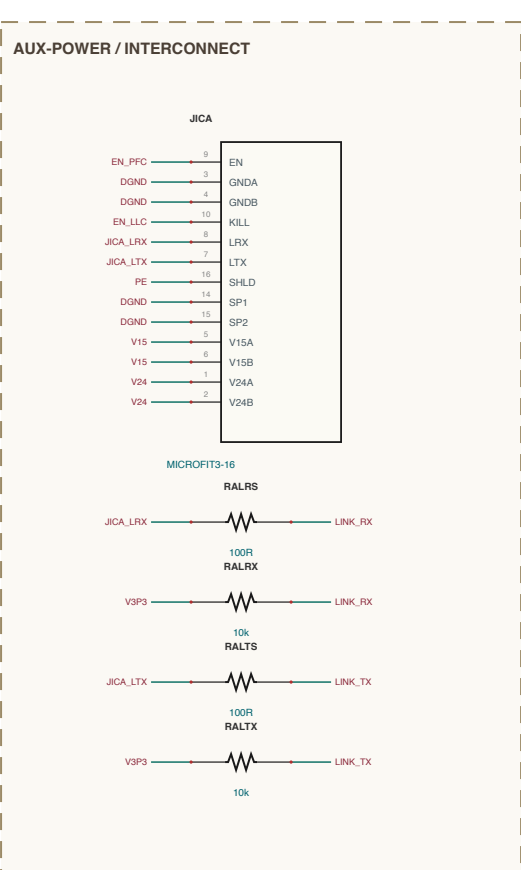
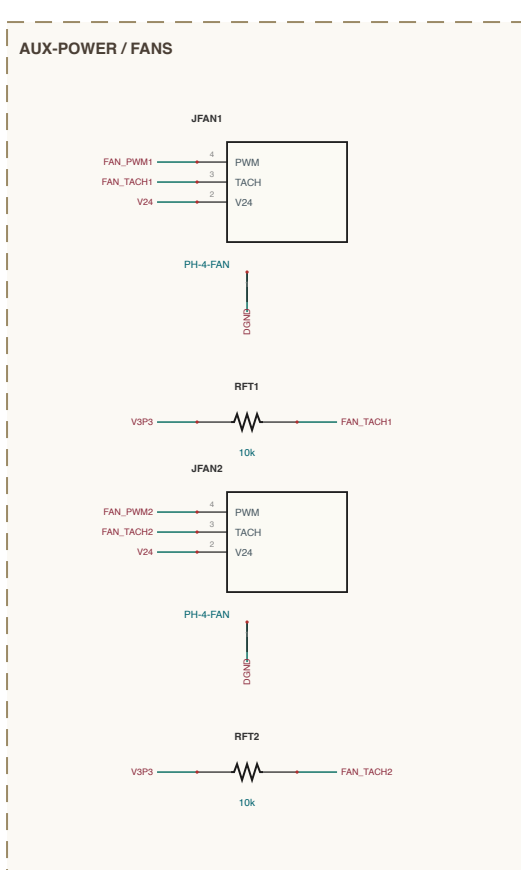
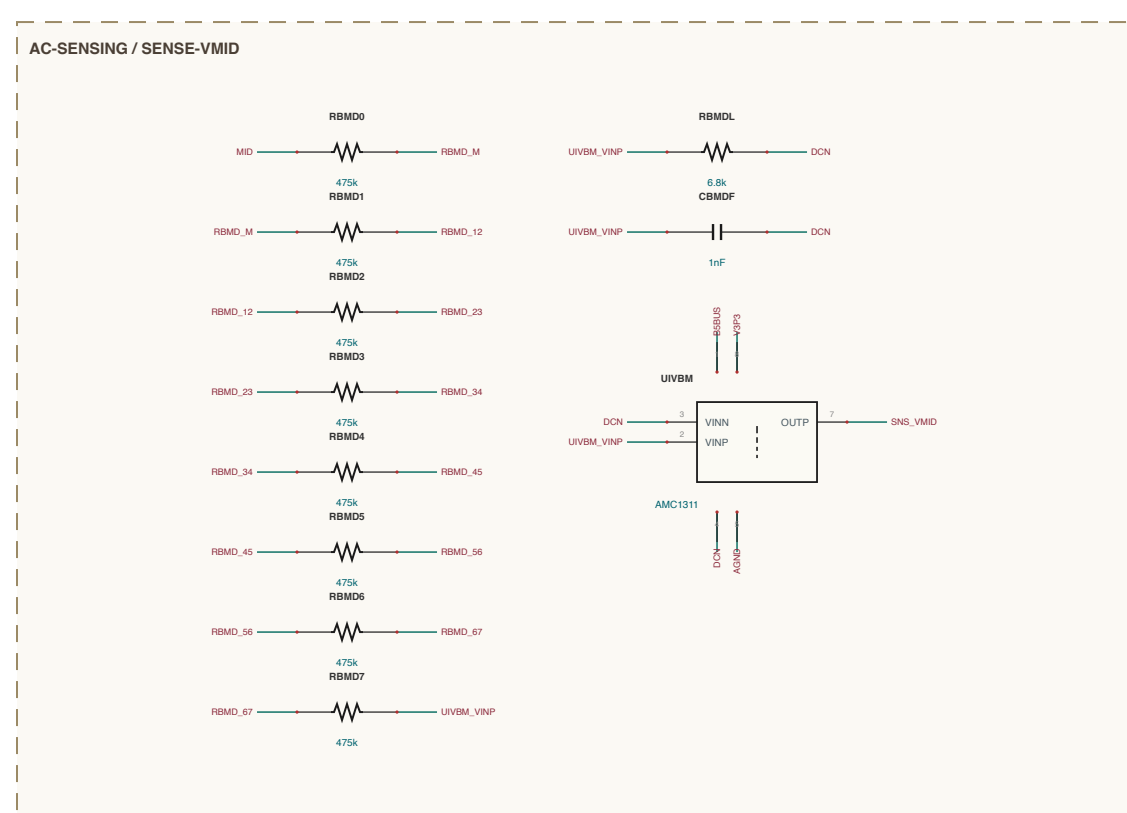
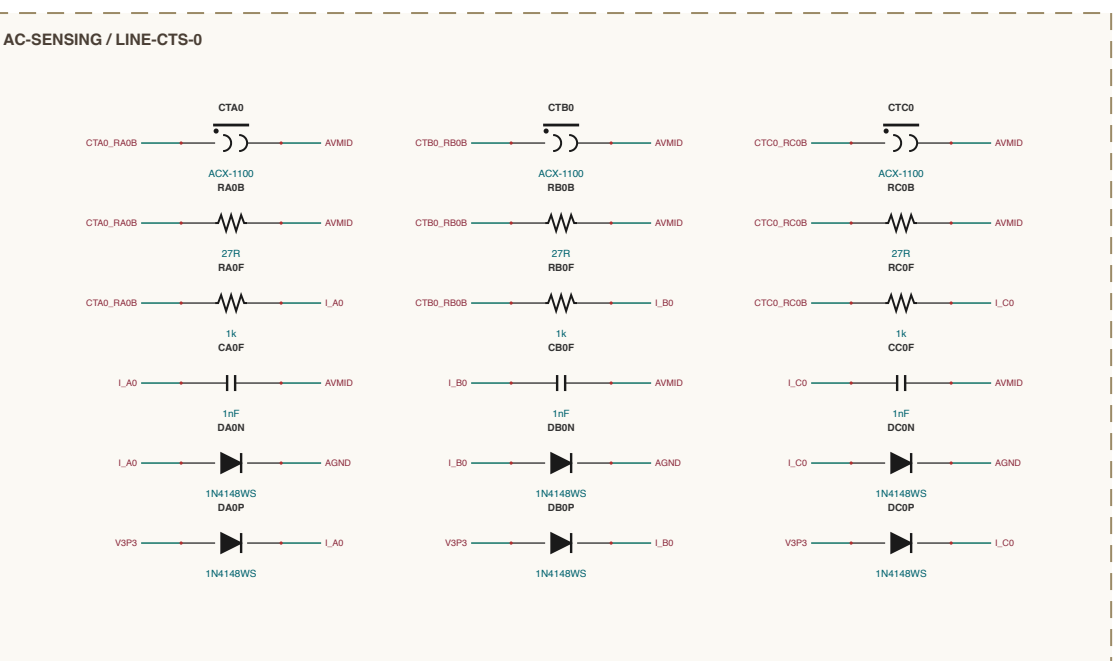
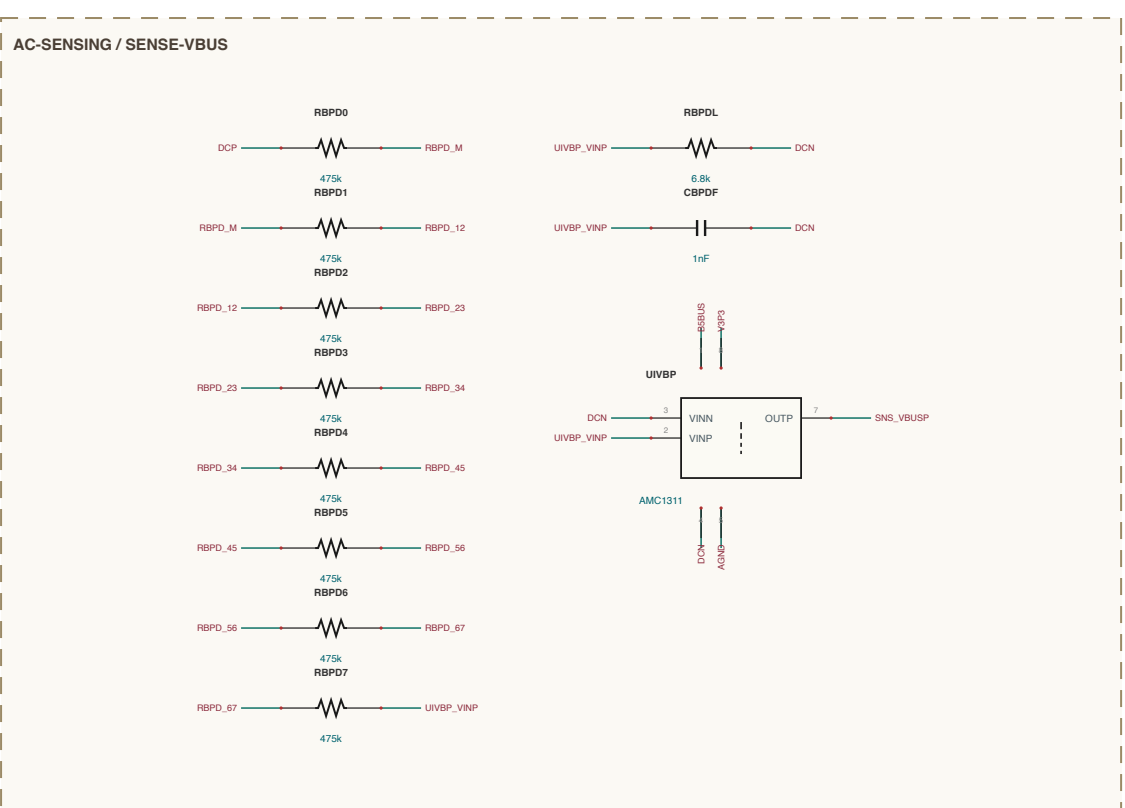
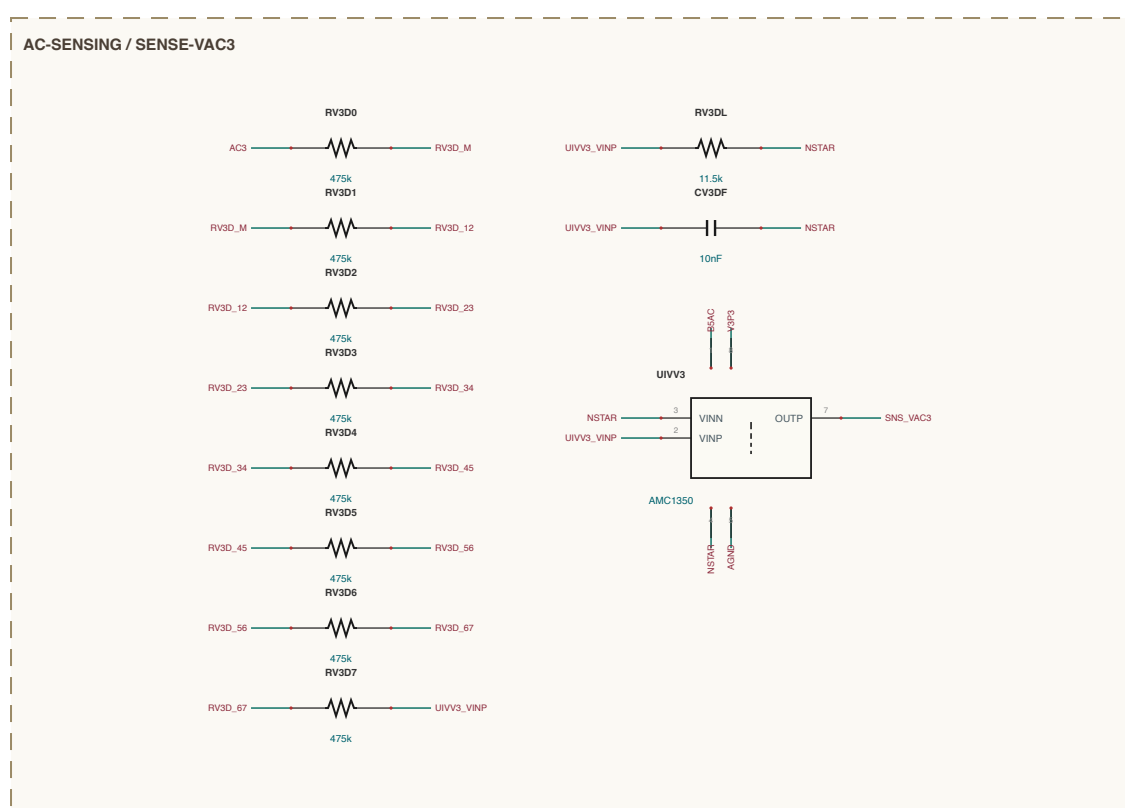
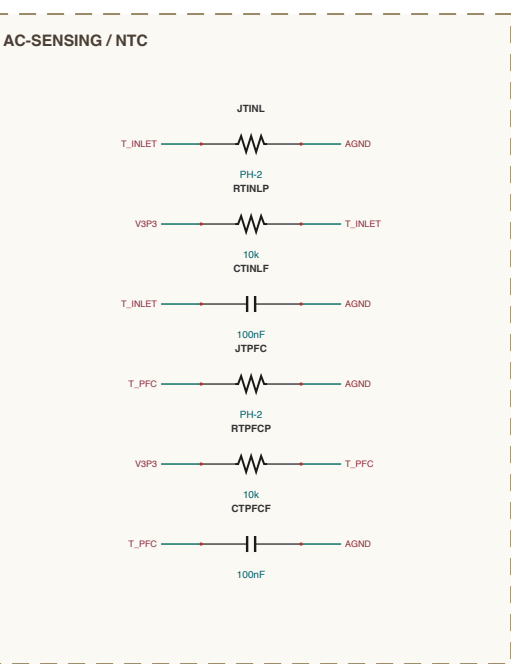
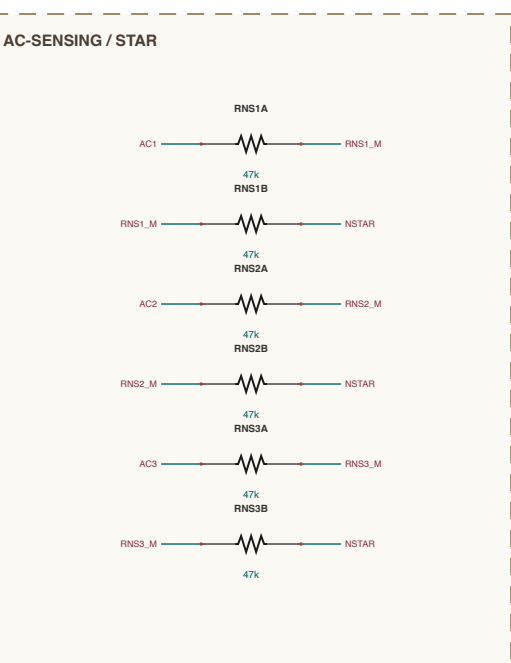
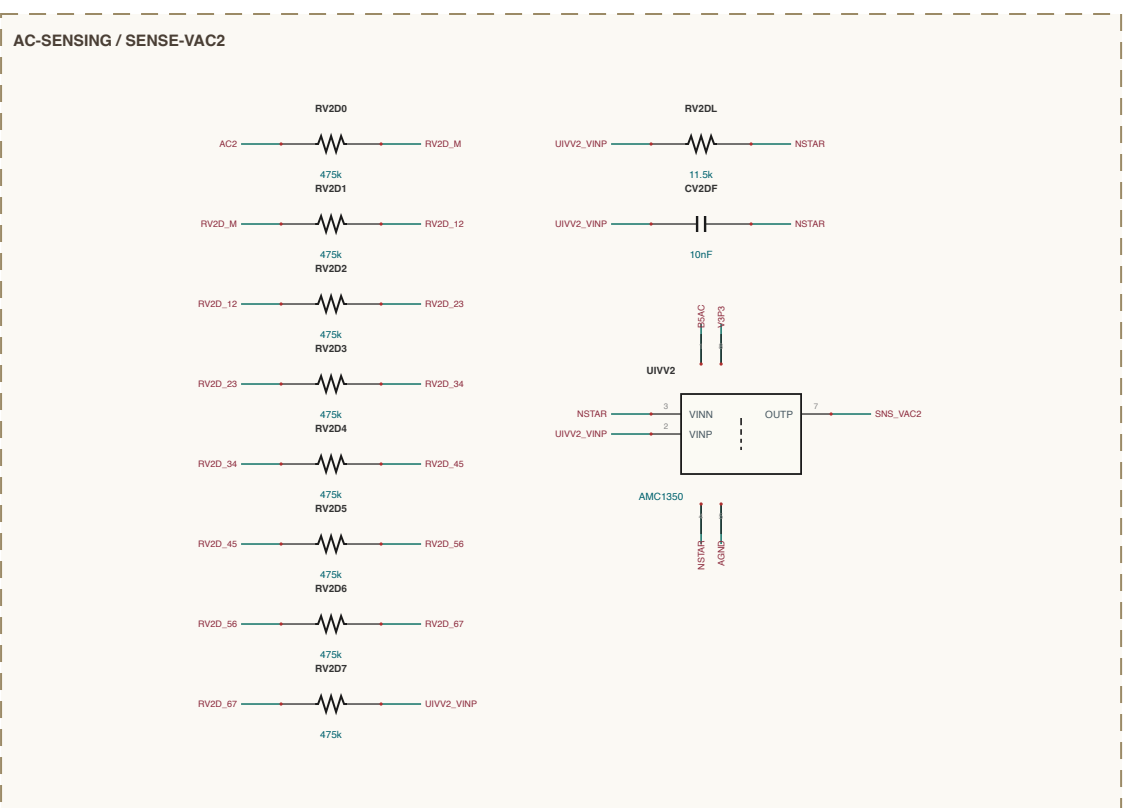
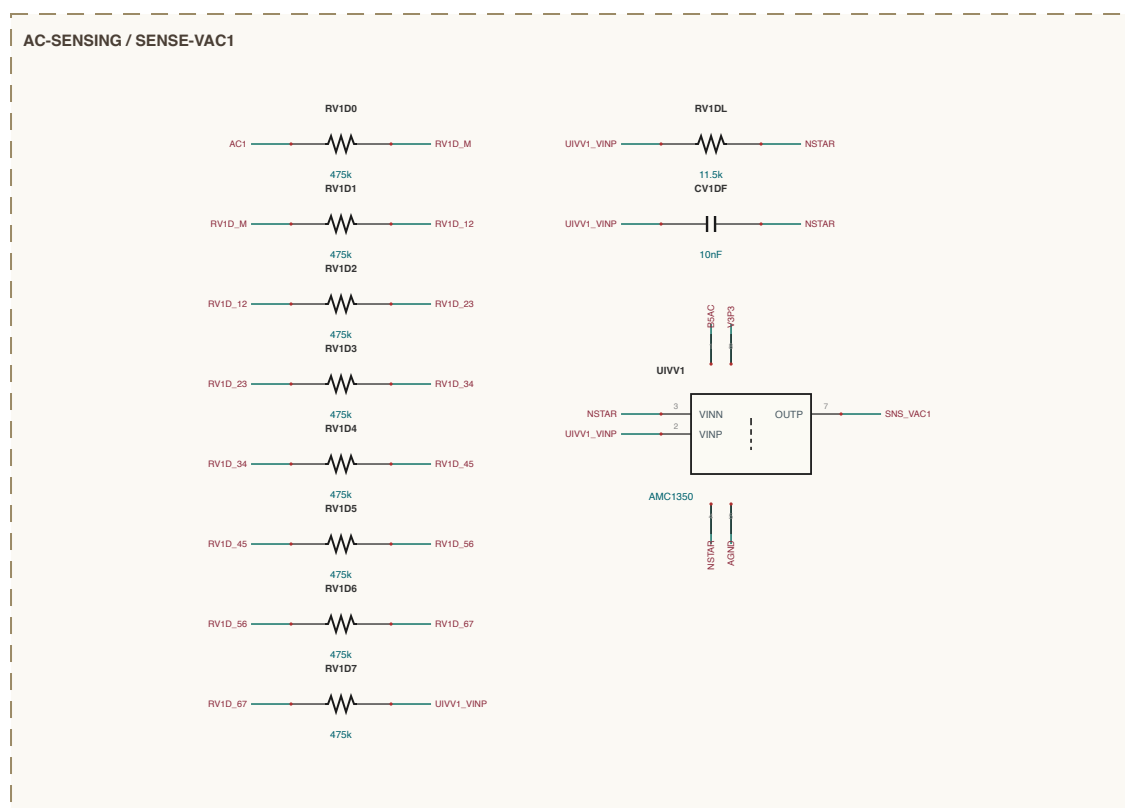
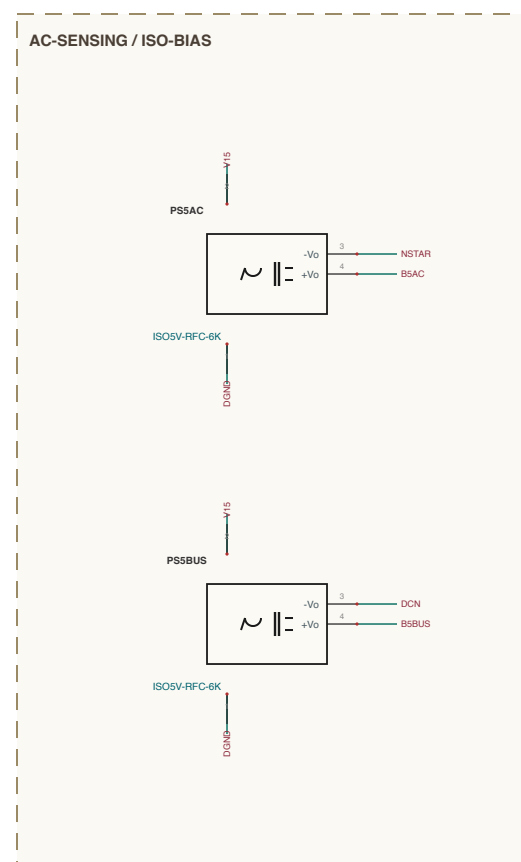




SHEET INDEX

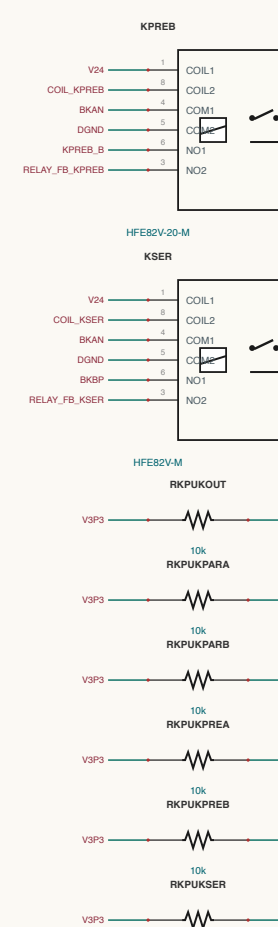
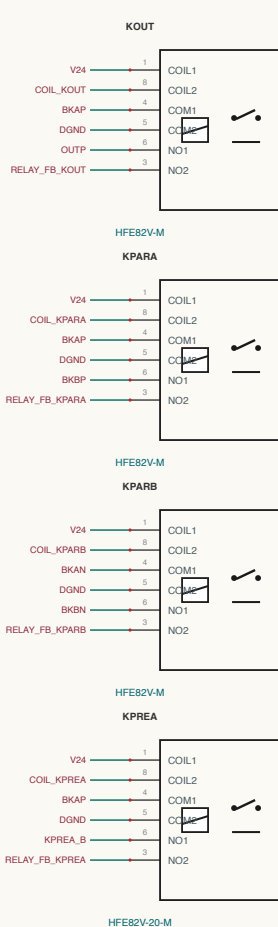
38 kW AC-DC (Line) - 1 of 3
AC-BENING - 8 sections
AUX POWER - 4 sections
CONTROL - 4 sections
REFILL GNE - 4 sections
DC-LINK - 3 sections
YINSHU PVC - 3 sections
rev D-3 - 27 sections - 271 components

NET NAMING

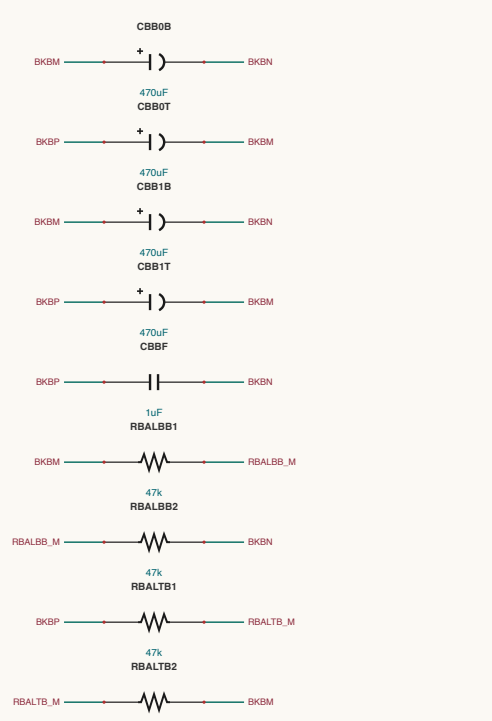
internal junctions are named for what they join:

- `Unctrl_0PMU_node` at that IC pin e.g. `VFIOA_VRP`
- `Resistor_M` midpoint of a series pair e.g. `IBALTA_M`
- `Resistor_vcrs` join between `Resistor1` and `Resistor2` e.g. `RFV1D_S1`
- all others are explicit design nets

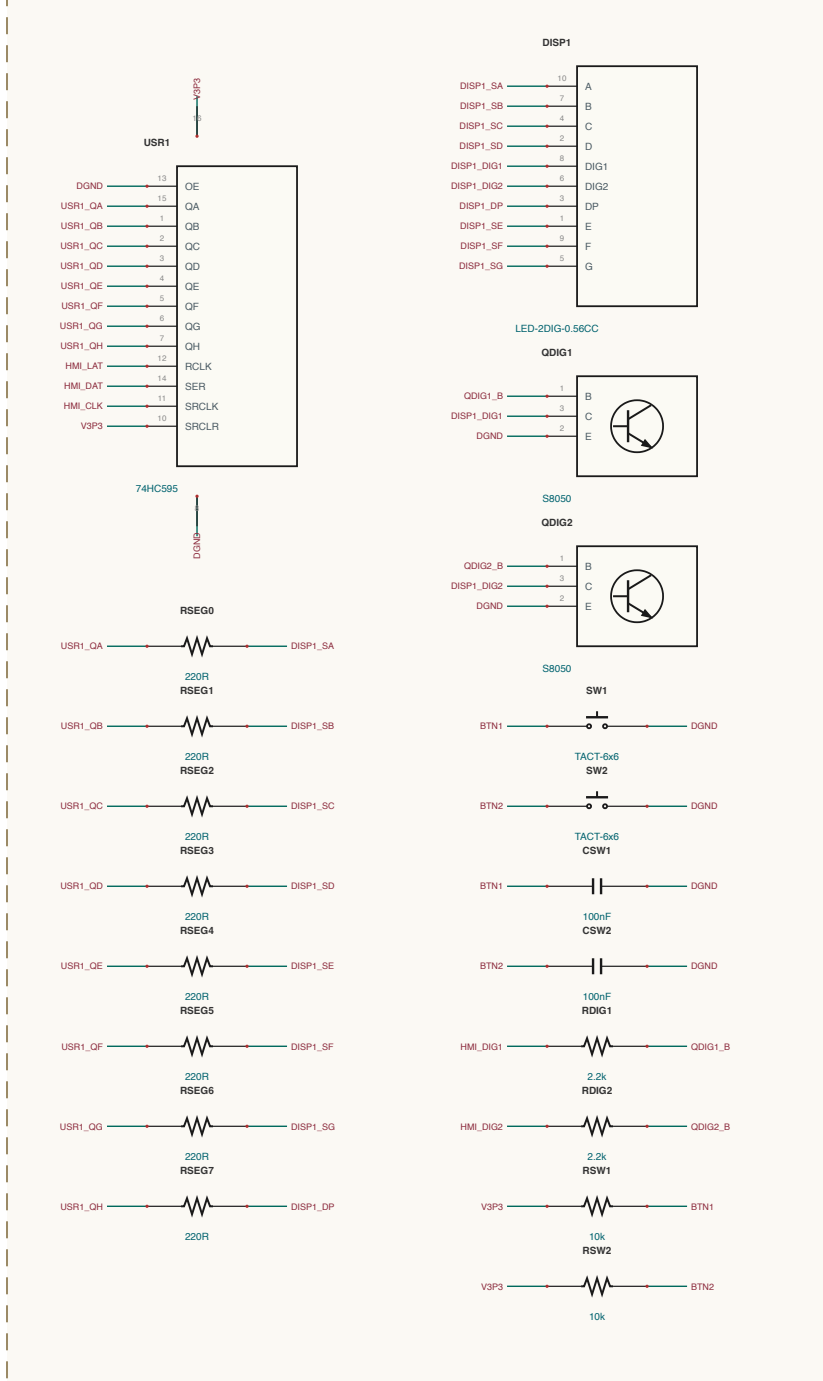
BANKS-SP / SP-MATRIX



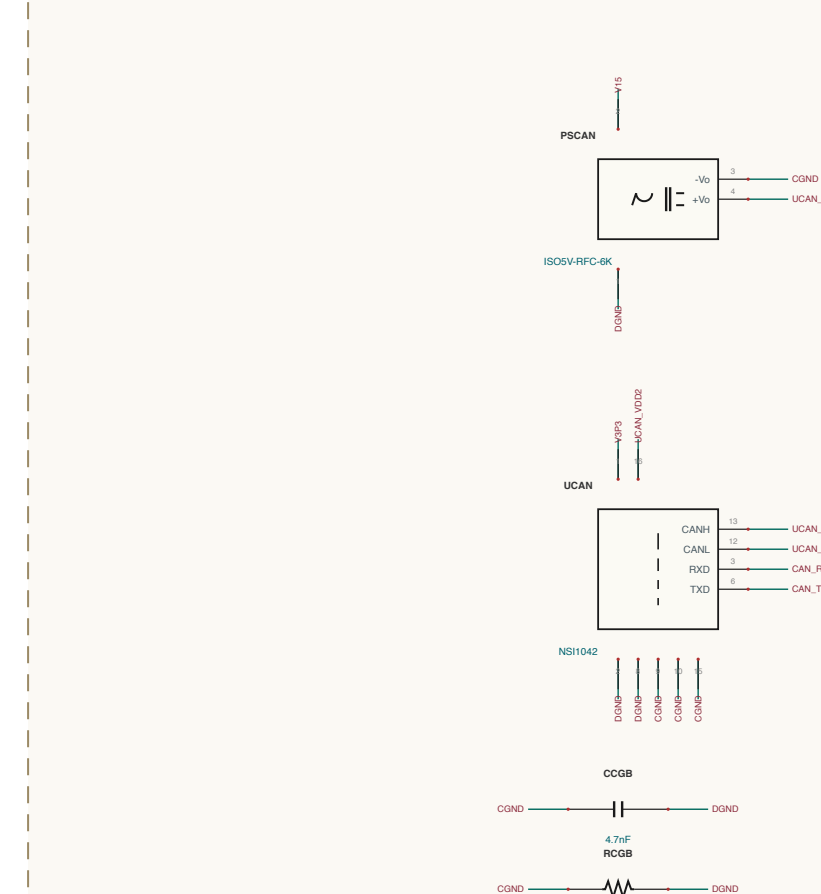
BANKS-SP / BANK-B



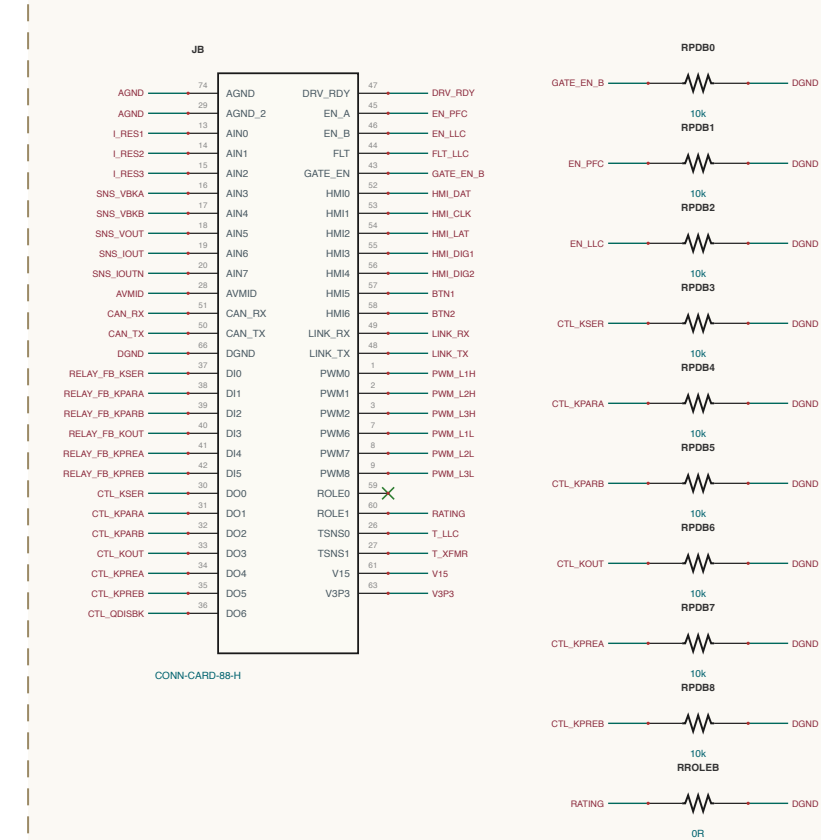
COMMS-HMI / HMI



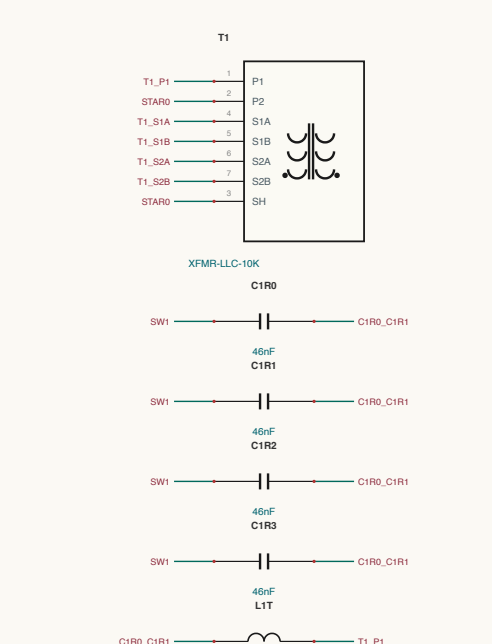
COMMS-HMI / CAN



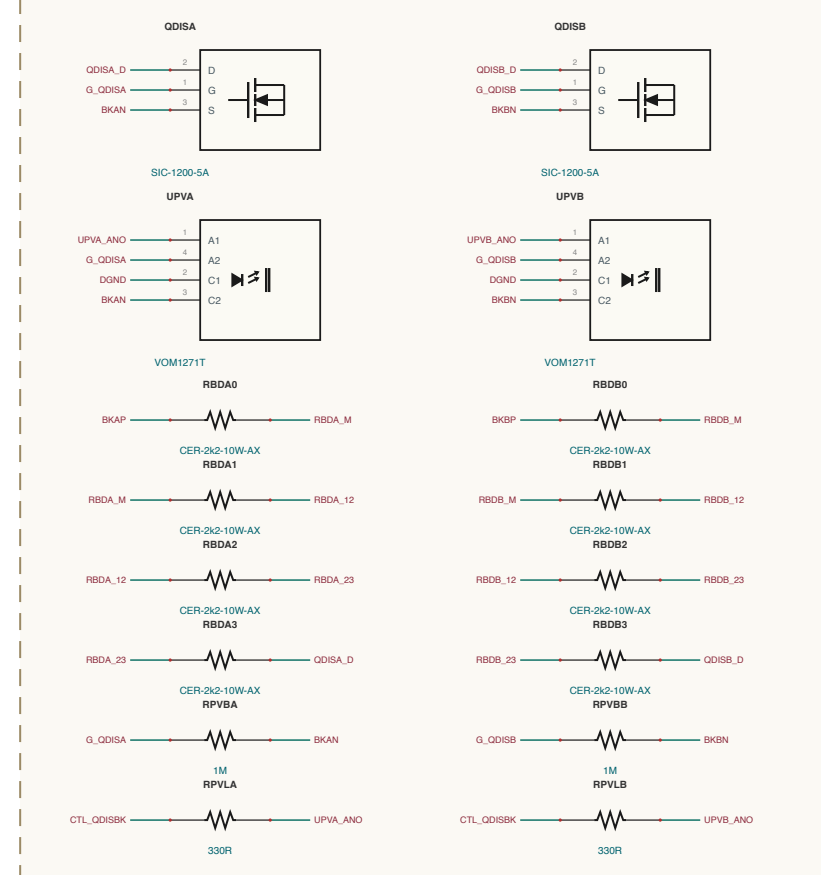
CONTROL / CARD-IP



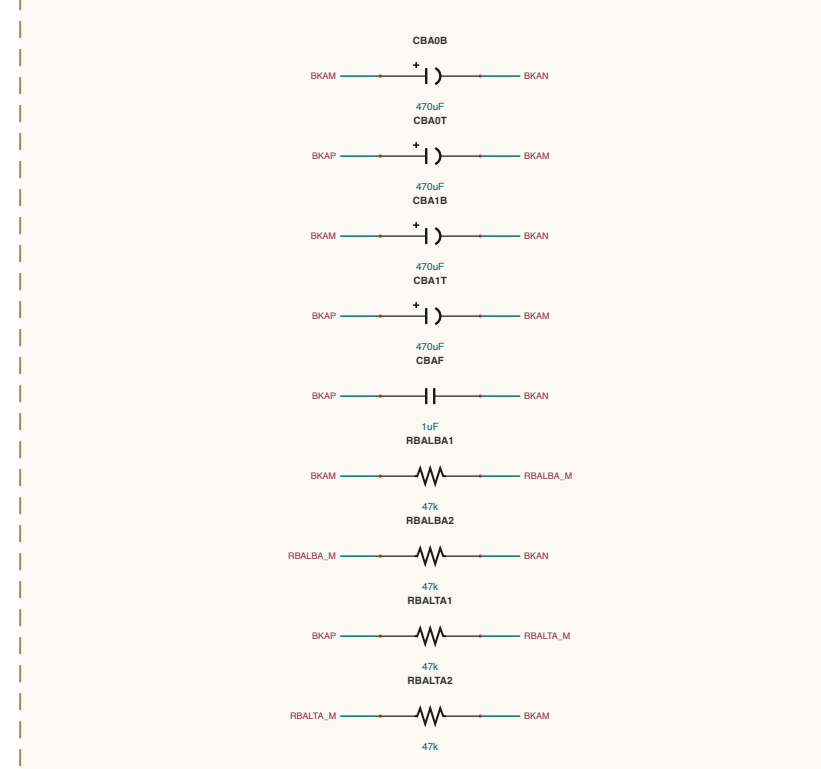
LLC-TANKS / TANK-1



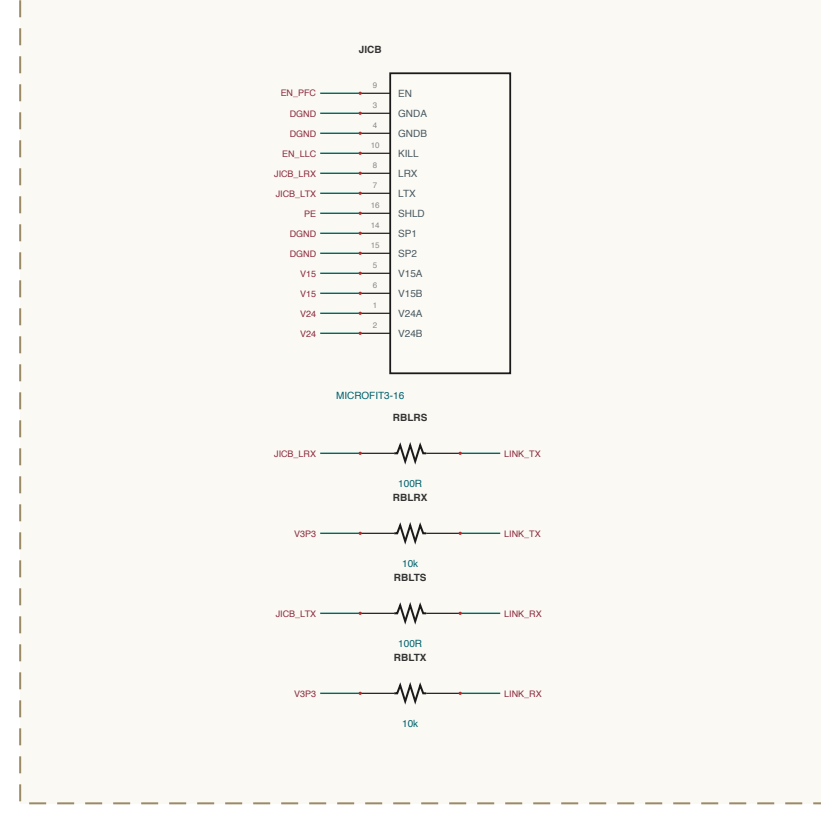
BANKS-SP / BLEEDERS



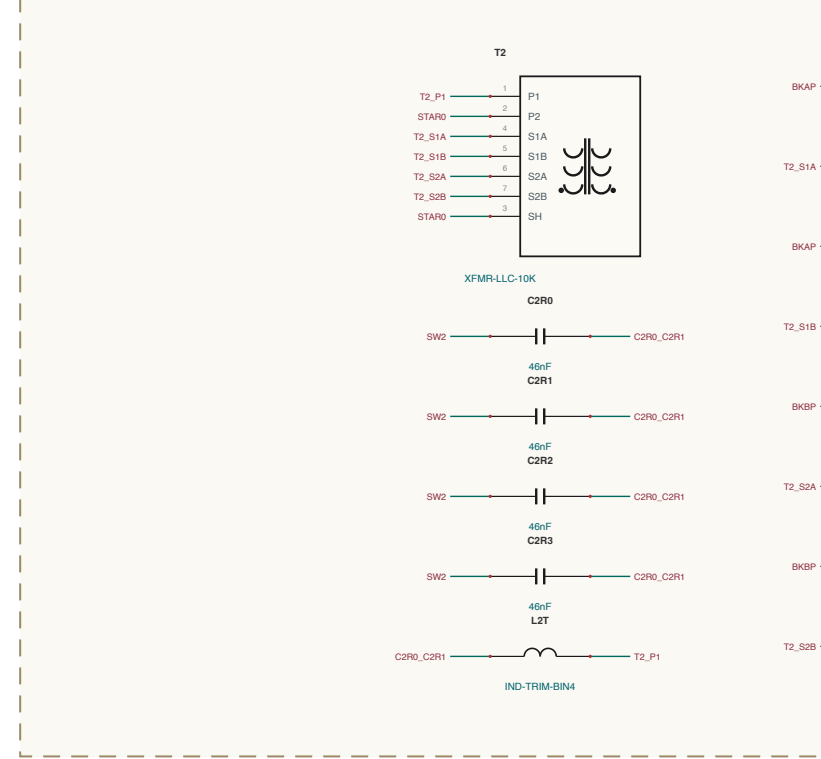
BANKS-SP / BANK-A



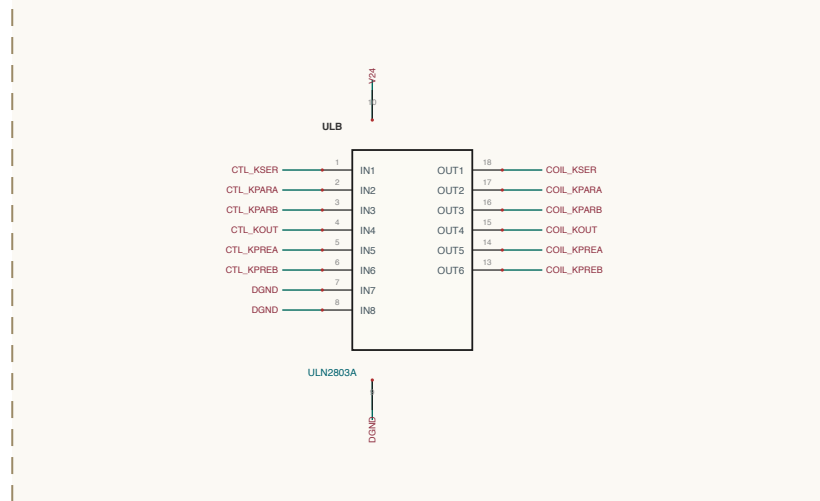
CONTROL / INTERCONNECT



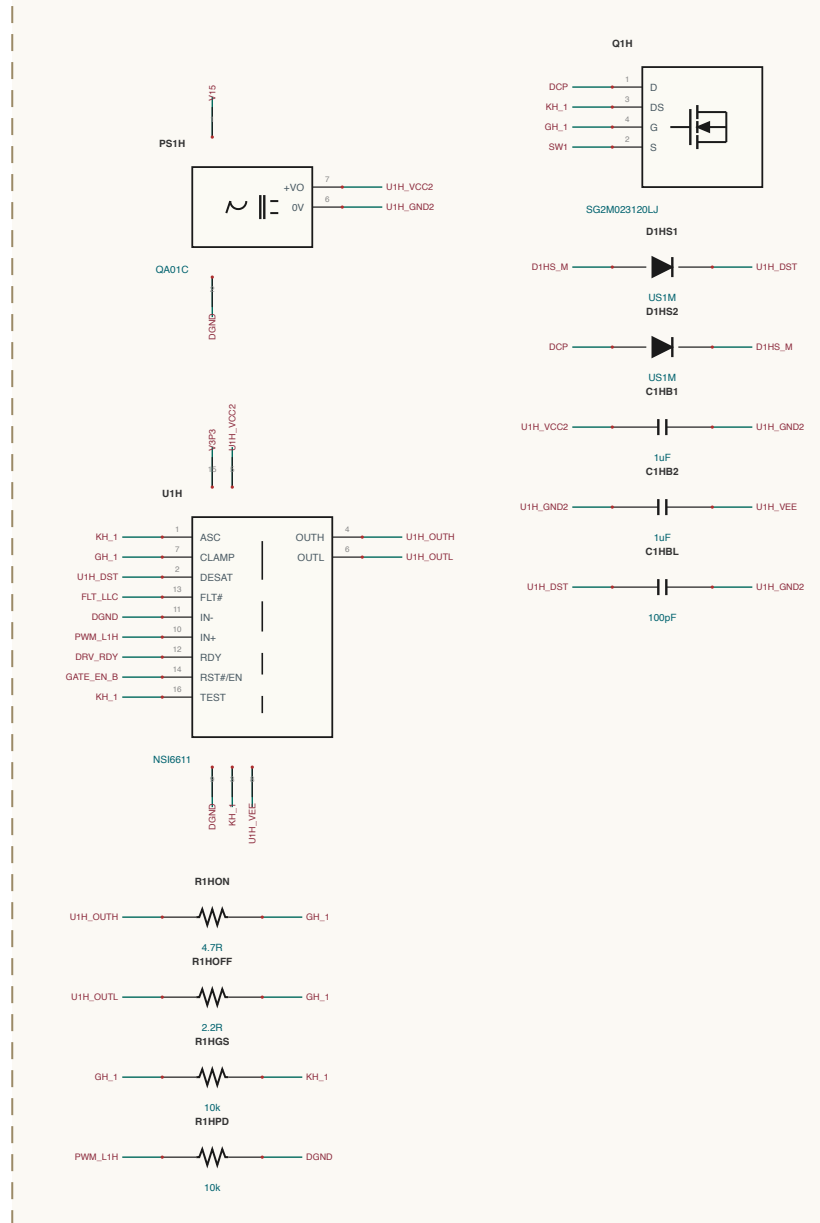
LLC-TANKS / TANK-2



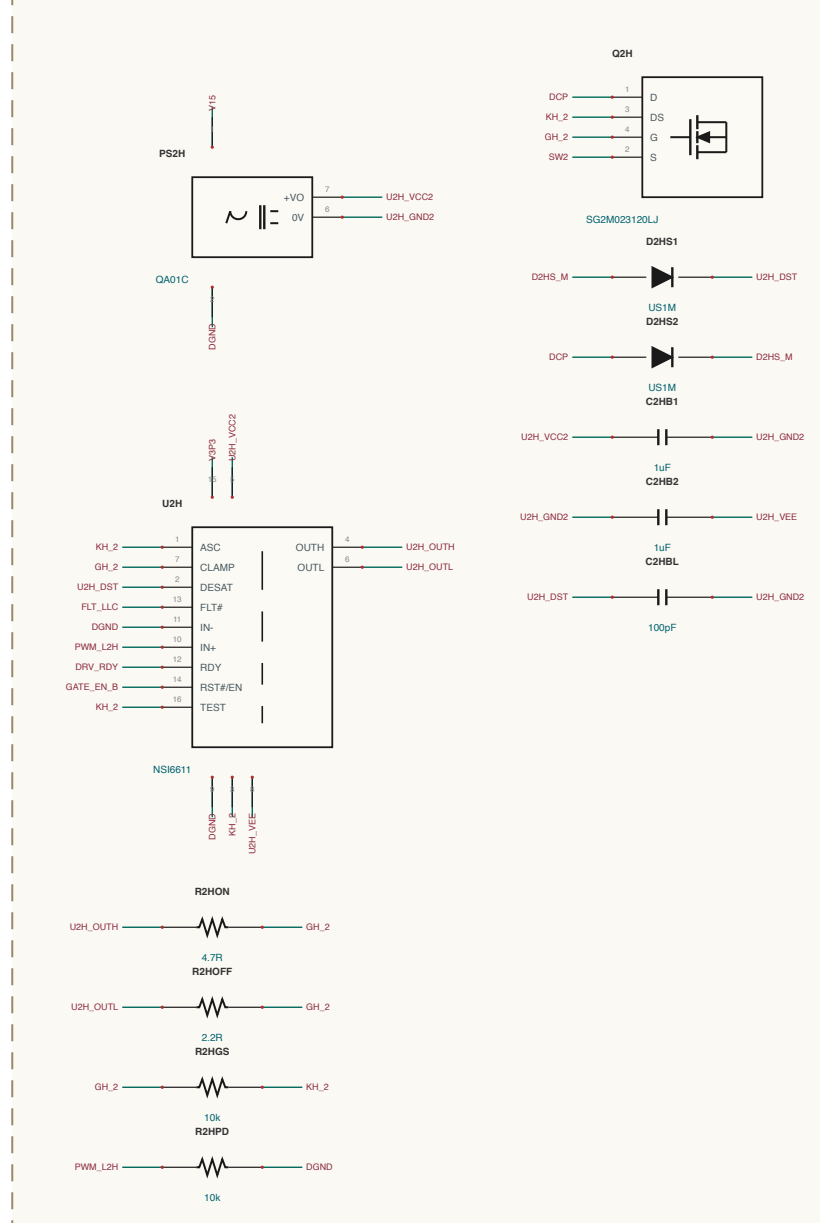
CONTROL / COL-DRIVER



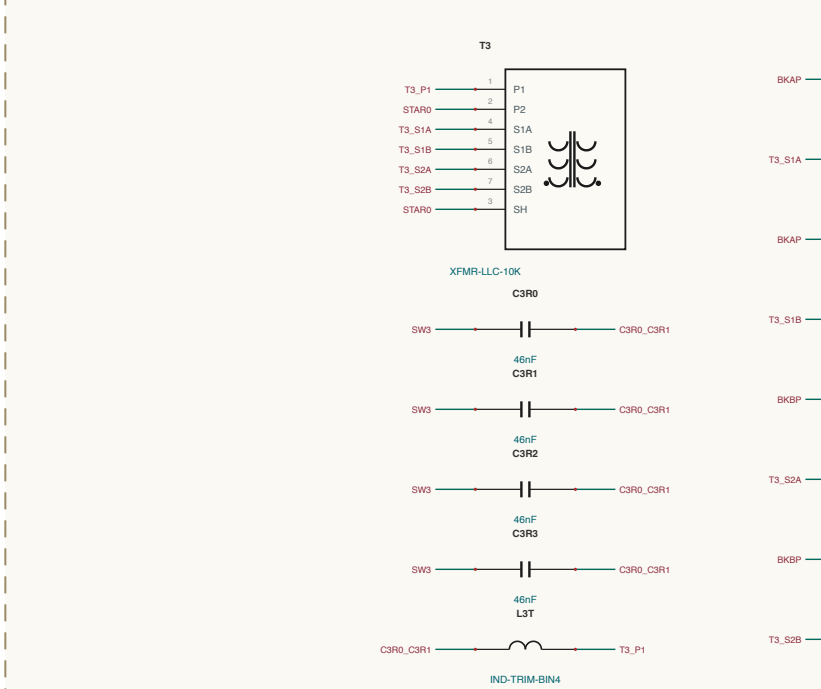
LLC-LEGS / LEG-1



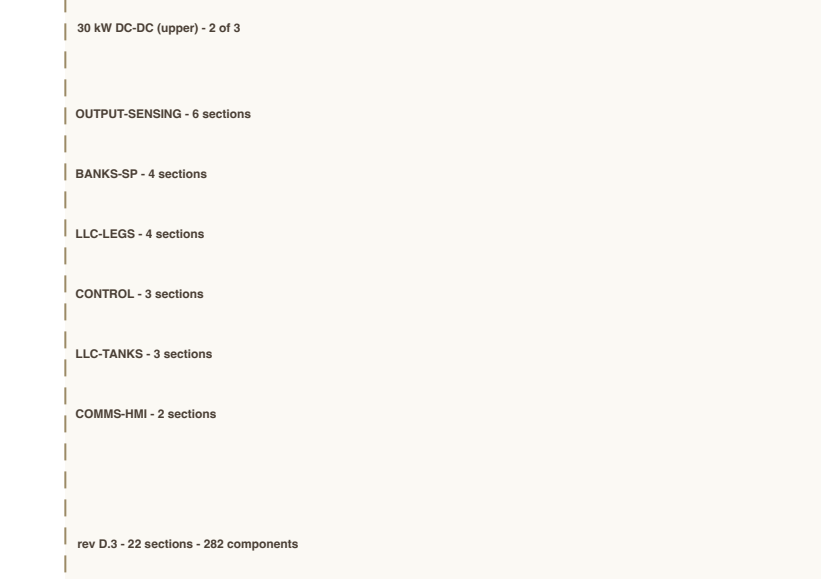
LLC-LEGS / LEG-2



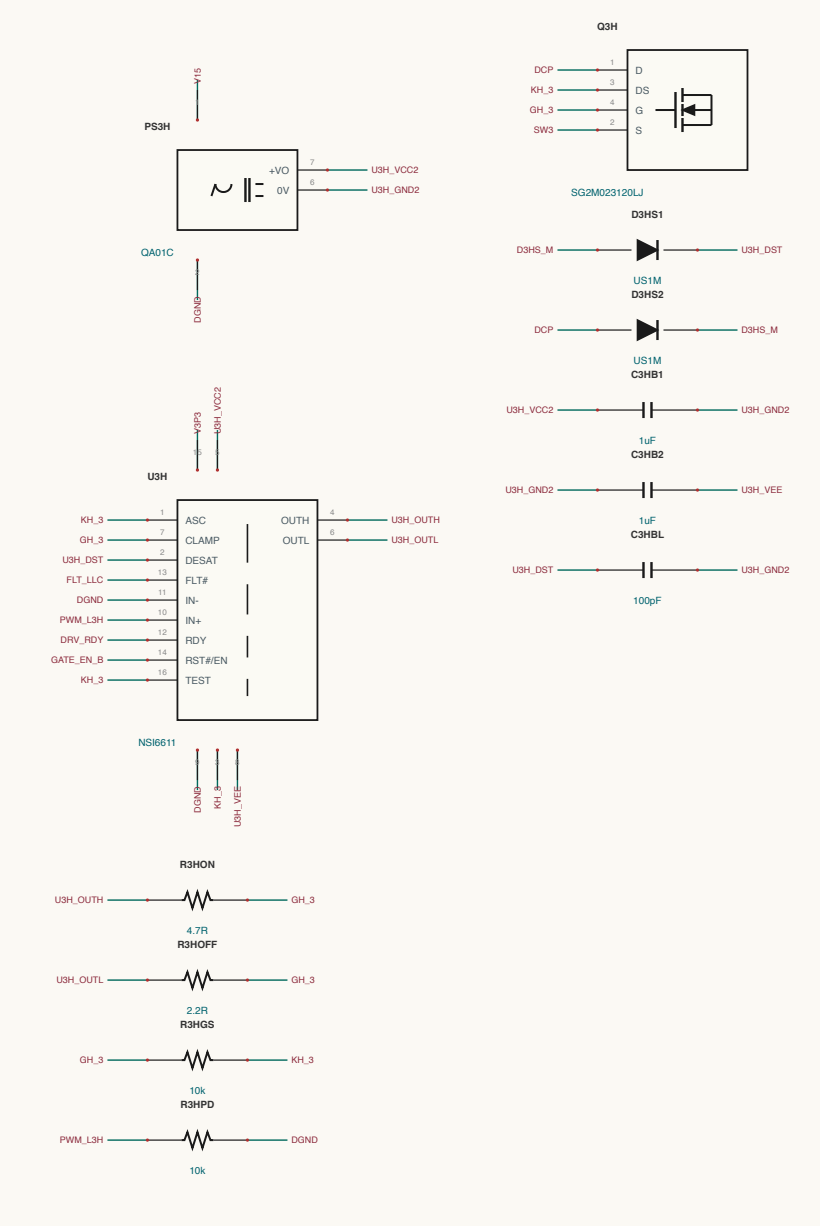
LLC-TANKS / TANK-3



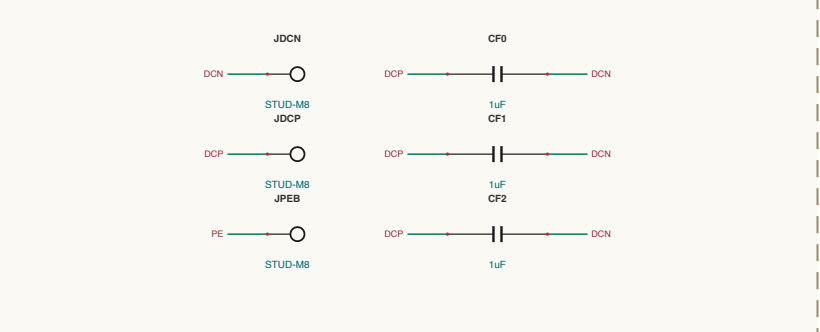
SHEET INDEX



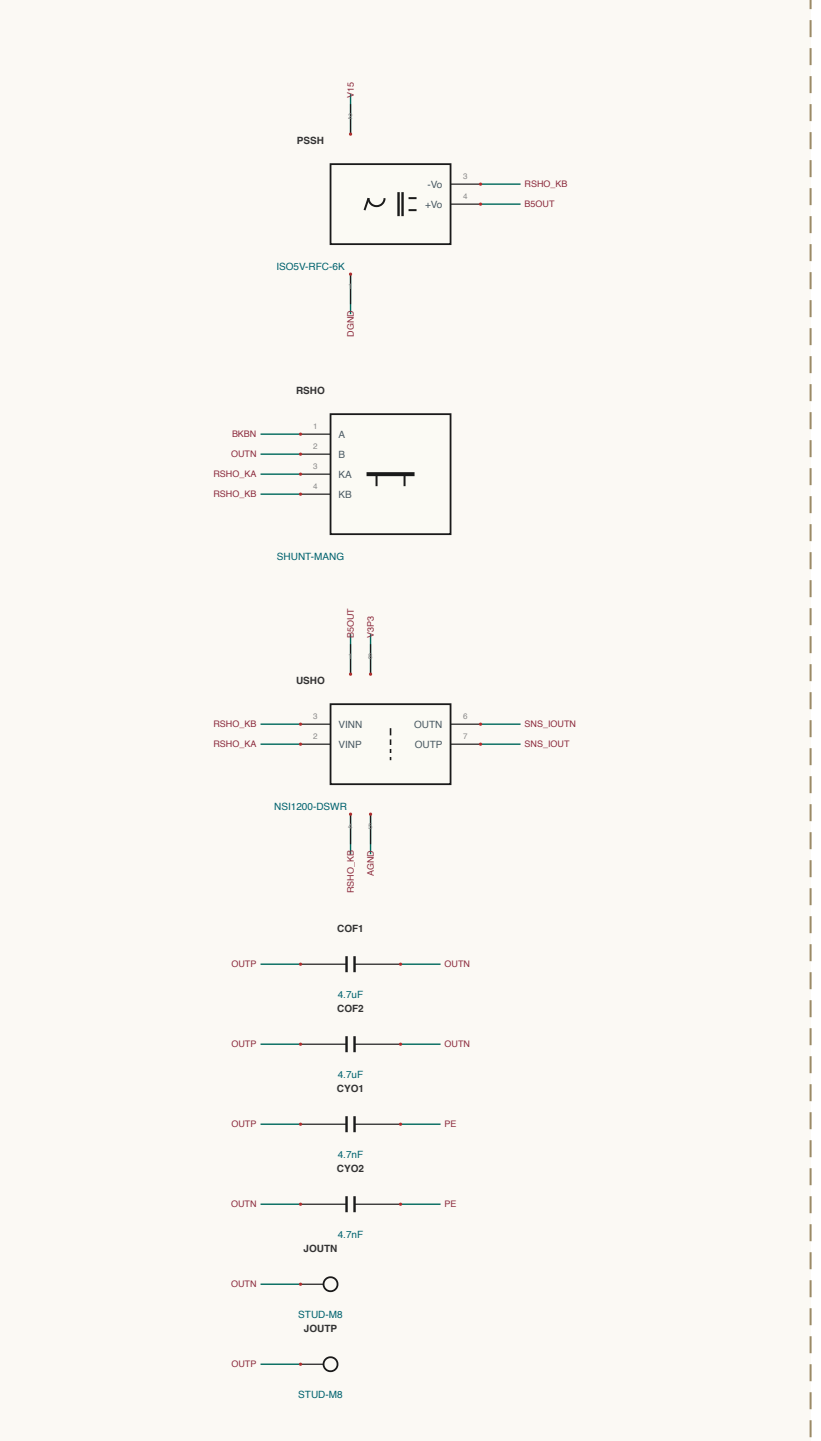
LLC-LEGS / LEG-3



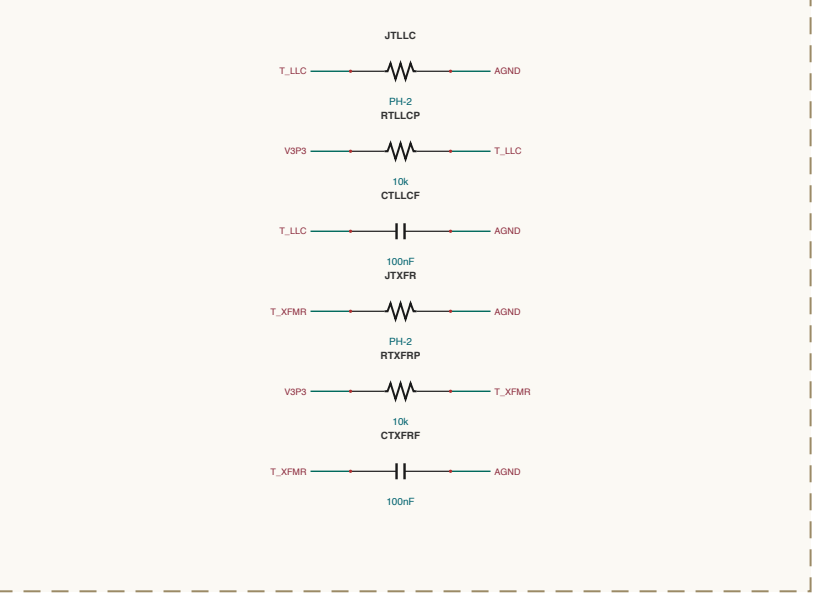
LLC-LEGS / BUS-IN



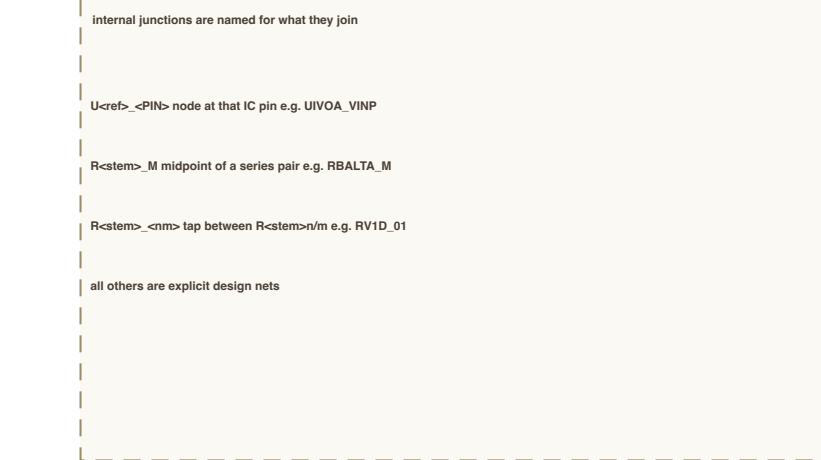
OUTPUT SENSING / OUTPUT



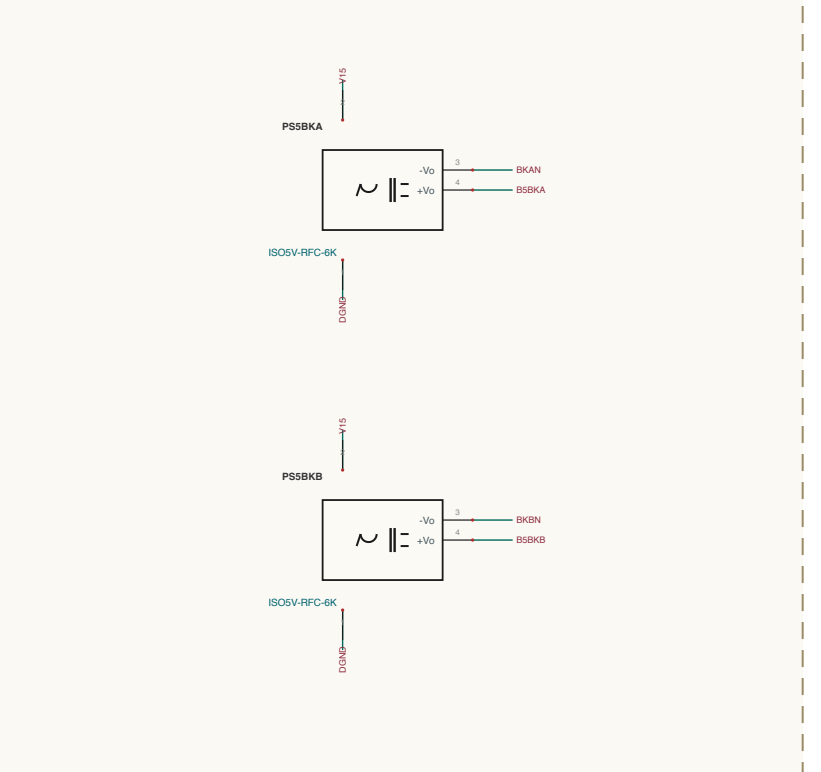
OUTPUT SENSING / ITC



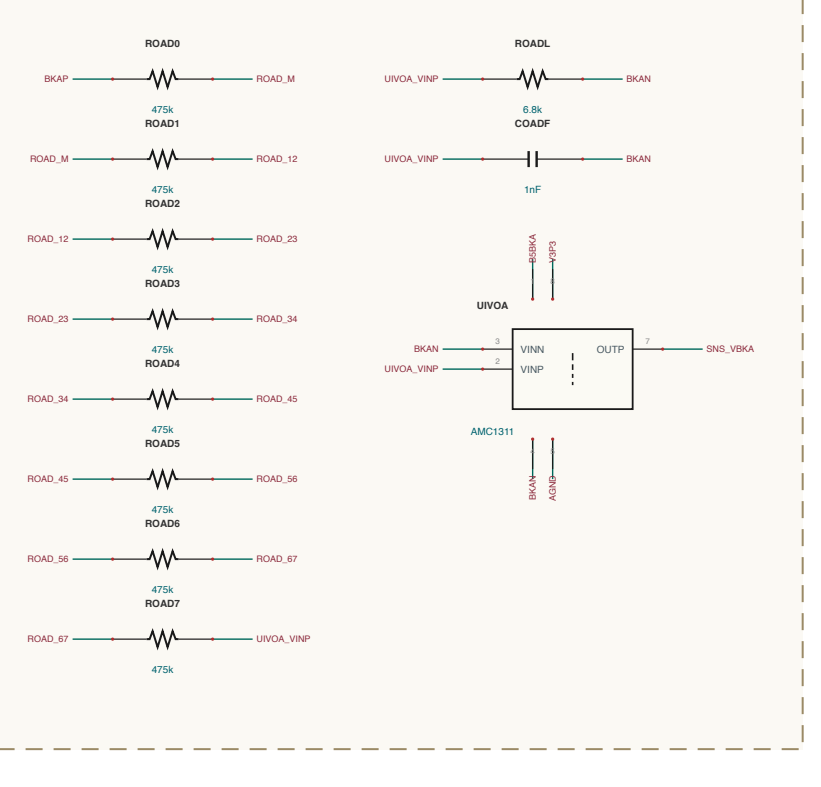
NET NAMING



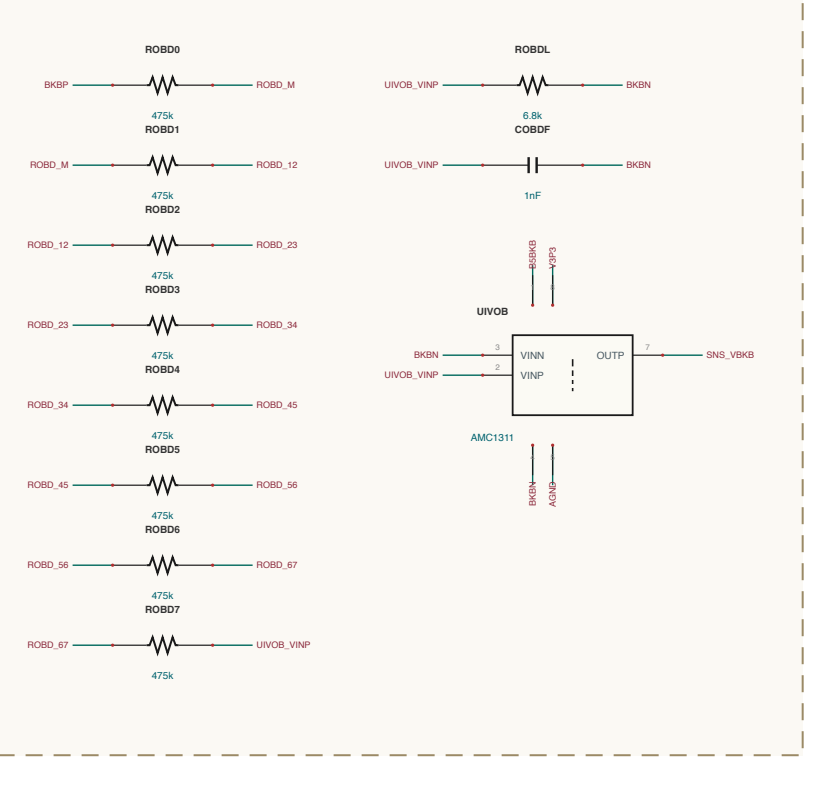
OUTPUT SENSING / BO-BIAS



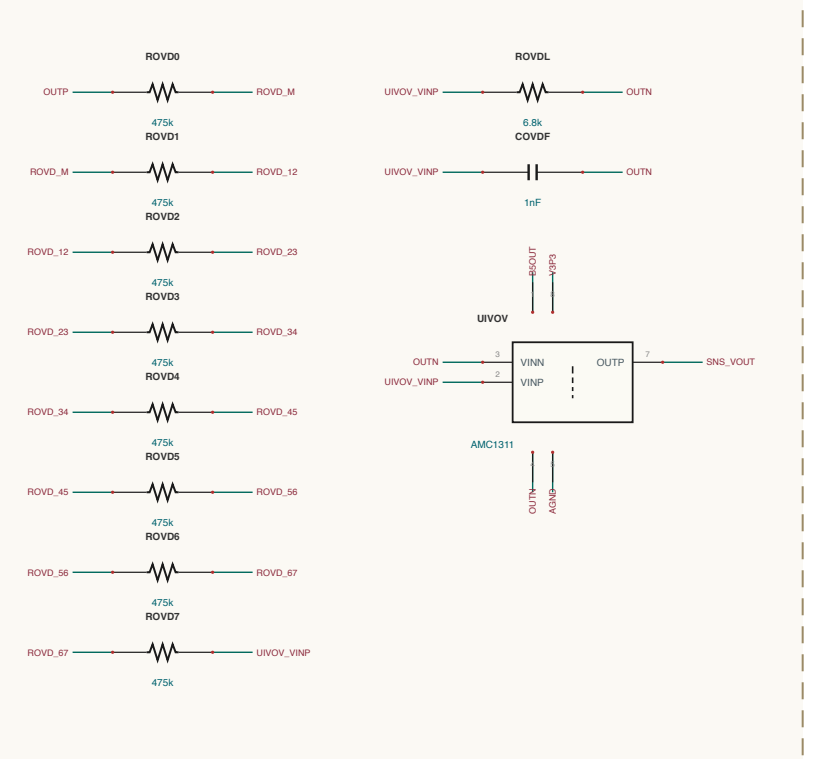
OUTPUT SENSING / SENSE-VBKA



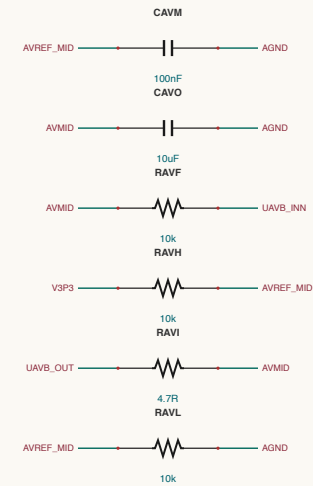
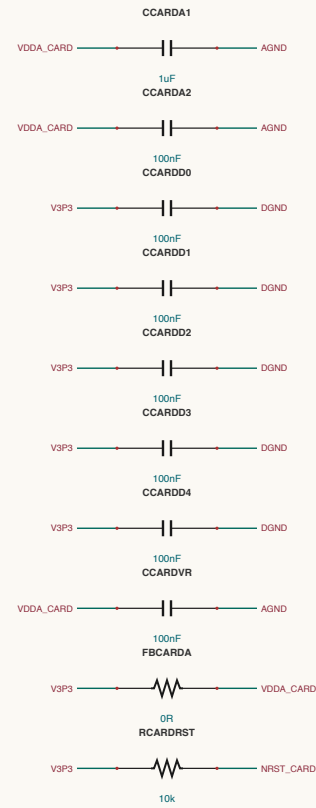
OUTPUT SENSING / SENSE-VBKB



OUTPUT SENSING / SENSE-VOUT



UCAR2					
V3P3	6	pin6	pin55	59	AIN7
WDI	11	pin11	pin56	58	AIN6
NRST_CARD	14	pin14	pin59	59	DO0
AIN9	15	pin15	pin60	60	DO1
AIN10	16	pin16	pin62	62	DO5
AIN8	17	pin17	pin63	63	DGND
AVMID	18	pin18	pin64	64	V3P3
AIN0	20	pin20	pin65	65	PWM5
AIN1	21	pin21	pin66	66	PWM11
AIN2	22	pin22	pin67	67	PWM4
DGND	23	pin23	pin68	68	PWM10
V3P3	24	pin24	pin69	69	PWM0
AIN11	25	pin25	pin70	70	PWM6
AIN12	26	pin26	pin71	71	PWM1
AIN3	27	pin27	pin72	72	PWM7
EN_B	28	pin28	pin73	73	D4
AIN4	29	pin29	pin74	74	DGND
AIN5	30	pin30	pin75	75	V3P3
TSNS0	32	pin32	pin76	76	SWDIO_CARD
TSNS1	33	pin33	pin77	77	SWCLK_CARD
DO6	34	pin34	pin78	78	D15
AGND	35	pin35	pin81	81	DRV_RDY
VDDA_CARD	36	pin36	pin82	82	EN_A
VDDA_CARD	37	pin37	pin84	84	HM15
ROLE1	38	pin38	pin85	85	D10
HM10	40	pin40	pin86	86	D11
HM11	41	pin41	pin87	87	LINK_TX
HM12	42	pin42	pin88	88	LINK_RX
D12	43	pin43	pin89	89	DO2
D13	44	pin44	pin90	90	ROLE0
HM13	45	pin45	pin91	91	HM16
FLT	47	pin47	pin92	92	CAN_RX
DGND	48	pin48	pin93	93	CAN_TX
V3P3	49	pin49	pin94	94	DO3
HM14	50	pin50	pin95	95	BOOT0_CARD
PWM2	51	pin51	pin96	96	DO4
PWM8	52	pin52	pin99	99	DGND
PWM3	53	pin53	pin100	100	V3P3
PWM9	54	pin54			

[illegible]

The diagram illustrates the wiring of the JSWDCARD module to the Raspberry Pi 4. The module's pins are connected to the Pi's pins as follows:

- SWCLK_CARD** (green) connects to **CLK** (red).
- SWDIO_CARD** (green) connects to **DIO** (red).
- NRST_CARD** (green) connects to **RST** (red).
- HDR-1x5-2.54** (blue) connects to **D009** (red).
- CCARDRST** (black) connects to **NRST_CARD** (green).

Additionally, a **100nF** capacitor is connected between **NRST_CARD** and ground, and a **10k** resistor is connected between **BOOT0_CARD** and ground.

The diagram illustrates three different pull-up configurations for a circuit:

- CFLT to DGND:** A green line labeled "CFLT" is connected to a red line labeled "DGND" through a capacitor labeled "1nF".
- RAGND to DGND:** A green line labeled "RAGND" is connected to a red line labeled "DGND" through a resistor labeled "0R".
- VOP3 to FLT:** A green line labeled "VOP3" is connected to a red line labeled "FLT" through a resistor labeled "4.7k".

The diagram shows two identical resistor components, RROLE0 and RROLE1, connected in parallel. Each resistor is represented by a zigzag line with a value of 10k. The left terminal of each resistor is connected to a green line labeled V3P3, and the right terminal is connected to a green line labeled GND.

The schematic diagram illustrates the internal components and connections of the UBKCARD module. The module is represented by a central box with four pins: 1 (BOOT), 2 (SW), 3 (EN), and 4 (FB). The connections are as follows:

- Pin 1 (BOOT):** Connected to a V15 voltage source.
- Pin 2 (SW):** Connected to a TPS54202 DC-DC converter.
- Pin 3 (EN):** Connected to a 10uF capacitor (CBKICARD) and a 45.3k resistor (RBKF2CARD).
- Pin 4 (FB):** Connected to a 10k resistor.
- Internal Connections:**
 - A 22uF capacitor (CBSTCARD) is connected between pins 3 and 4.
 - A 100nF capacitor (LBKCARD) is connected between pins 2 and 3.
- External Connections:**
 - A V15 voltage source is connected to the BOOT pin.
 - A DGND ground is connected to the SW pin.

JCARD				
AGND	74	AGND	47	DRV_RDY
AGND	29	AGND_2	45	EN_A
AIN0	13	AIN0	46	EN_B
AIN1	14	AIN1	44	FLT
AIN2	15	AIN2	43	GATE_EN
AIN3	16	AIN3	52	HM0
AIN4	17	AIN4	53	HM1
AIN5	18	AIN5	54	HM2
AIN6	19	AIN6	55	HM3
AIN7	20	AIN7	56	HM4
AIN8	21	AIN8	57	HM5
AIN9	22	AIN9	58	HM6
AIN10	23	AIN10	49	LINK_RX
AIN11	24	AIN11	48	LINK_TX
AIN12	25	AIN12	1	PWM0
AVMID	28	AVMID	2	PWM1
CAN_RX	51	CAN_RX	3	PWM2
CAN_TX	50	CAN_TX	4	PWM3
DGND	66	DGND	5	PWM4
DIO	37	DIO	6	PWM5
DIO1	38	DIO1	7	PWM6
DIO2	39	DIO2	8	PWM7
DIO3	40	DIO3	9	PWM8
DIO4	41	DIO4	10	PWM9
DIO5	42	DIO5	11	PWM10
DO0	30	DO0	12	PWM11
DO1	31	DO1	59	ROLE0
DO2	32	DO2	80	ROLE1
DO3	33	DO3	26	TSNS0
DO4	34	DO4	27	TSNS1
DO5	35	DO5	61	V15
DO6	36	DO6	63	VSP3

CONN-CARD-89-R

The schematic diagrams show the following components and their connections:

- 1nF C5FCARD**: Connected between V3P3 and DGNND.
- 100nF CWDCCARD**: Connected between USUPCARD_CWD and DGNND.
- 1nF R5NLCARD**: Connected between EN_A and DGNND.
- 100k R5NRCARD**: Connected between EN_B and DGNND.
- 100k R5GPDCCARD**: Connected between GATE_EN and DGNND.
- 10k R5RDYCARD**: Connected between V3P3 and DRV_RDY.
- 10k R5WPUCARD**: Connected between V3P3 and WDO_CARD.